

SPATIO TEMPORAL VARIABILITY OF URBAN HEAT
ISLAND THROUGH REMOTE SENSING AND
MACHINE
LEARNING TO ASSESS SOCIO-ECOLOGICAL
EFFECTS
ON SUSTAINABLE DEVELOPMENT

A PROJECT REPORT

Submitted by

MUBEENA MIYAMMAL M

MOHAMED BAZIL SM

MOHAMED ABDUL LATHEEF M

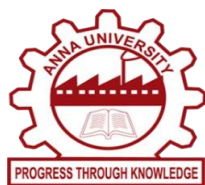
MANOSELVAN R

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Certified that this project report “SPATIO TEMPORAL
VARIABILITY OF URBAN HEAT ISLAND THROUGH REMOTE
SENSING AND MACHINE LEARNING TO ASSESS SOCIO
ECOLOGICAL EFFECTS ON SUSTAINABLE DEVELOPMENT”
is the bonafide work of “MUBEENA MIYAMMAL M,
MOHAMED BAZIL SM, MOHAMED ABDUL LATHEEF M,
MANOSELVAN R” who carried out the project work under
my supervision.

Dr. S. ADISHKUMAR,
M.E., Ph.D.,
HEAD OF THE DEPARTMENT,
Assistant professor,
Department of Civil
Engineering,
Anna University Regional
Campus, Tirunelveli – 627

Dr. S. ADISHKUMAR, M.E.,
Ph.D.,
SUPERVISOR,
HEAD OF THE DEPARTMENT,
Assistant Professor,
Department of Civil
Engineering,
Anna University Regional

007.

Campus, Tirunelveli – 627
007.

Submitted for the VIVA-VOCE Examination held on_____

INTERNAL EXAMINER
EXTERNAL EXAMINER

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Mubeena Miyammal M

Mohamed Bazil SM

Mohamed Abdul Latheef M

Manoselvan R

ABSTRACT

This research presents a comprehensive geospatial methodology for assessing urban environmental dynamics through the integration of remote sensing and ecological modeling. Utilizing Google Earth Engine (GEE), the study leverages Landsat-8 multi-spectral and thermal (TIRS) data spanning the period from 2016 to 2025. The workflow begins with rigorous data pre-processing, including cloud masking, radiometric scaling, and annual aggregation, to ensure high-fidelity temporal analysis. The core analytical framework involves the extraction of critical biophysical parameters and spectral indices. Land Surface Temperature (LST) is derived through thermal band conversion, while indices such as NDVI (vegetation), NDWI (water), and NDBI (built-up) are calculated to characterize surface composition. These metrics are further synthesized into the Fractional Vegetation Cover (FVC) and used to perform Land Use Land Cover (LULC) classification. The study advances into a multi-dimensional Thermal and Ecological Analysis, quantifying the Urban Heat Island (UHI) effect and the Urban Thermal Field Variance Index (UTFVI). These thermal indicators are cross-referenced with ecological health metrics, specifically the Heat Health Risk Index (HHRI)—which evaluates hazard, exposure, and vulnerability—and the Crop Water Stress Index (CWSI) to monitor vegetation transpiration and moisture deficits. Final outputs are subjected to statistical

validation and temporal trend tracking to ensure model accuracy. This integrated approach provides a robust scalable model for urban planners and environmental scientists to mitigate climate-related risks and optimize ecological resilience in rapidly urbanizing landscapes.

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LIST OF ABBREVIATIONS

ARCGIS	Aeronautical Reconnaissance Coverage System	Geographic Information
DEM	Digital Elevation Model	
GEE	Google Earth Engine	
GIS	Geographic Information System	
IMD	Indian Meteorological Department	
LULC	Land Use and Land Cover	
ML	Machine Learning	
MS-EXCEL	Microsoft Excel	
SRTM	Shuttle Radar Topography Mission	
SWIR	Short Wave Infrared	
NDVI	Normalized Difference Vegetation Index	
NDWI	Normalized Difference Water Index	
NDBI	Normalized Difference Builtup Index	
LST	Land Surface Temperature	

FVC	Fractional Vegetation Cover
UTFVI	Urban Thermal Field Variance Index
UHI	Urban Heat Island
CWSI	Crop Water Stress index
HHRI	Heat Health Risk Index

CHAPTER 1

INTRODUCTION

1.1 GENERAL

The global landscape is undergoing a radical transition as rural areas are rapidly converted into urban centers. This transformation is characterized by the replacement of natural pervious surfaces with engineered materials like asphalt and concrete. While urbanization drives economic growth, it simultaneously disrupts the local energy balance, leading to the "Urban Heat Island" (UHI) effect. This study focuses on Thoothukudi, a prominent industrial coastal city in Tamil Nadu, to analyze its thermal evolution over the decade spanning 2016 to 2025. By integrating satellite remote sensing with ecological modeling, this research aims to quantify the relationship between intensive industrial land-use changes and environmental heat stress, providing a scientific basis for sustainable urban cooling and coastal climate resilience.

1.2 URBANIZATION AND ENVIRONMENTAL CHALLENGES

The rapid transition from rural to urban landscapes represents one of the most significant anthropogenic shifts of the 21st century. While urbanization is a primary driver of economic prosperity and modernization, it introduces profound structural changes to the

earth's surface, leading to a host of environmental challenges that threaten long-term climate resilience.

1.2.1 Global Trends in Urban Expansion

In the 21st century, more than half of the world's population resides in urban areas, a trend expected to reach 68% by 2050. In the context of Thoothukudi, this expansion is driven by its status as a major "Port City" and industrial hub. Such growth often occurs horizontally, consuming salt pans, agricultural land, and scrub forests. In rapidly developing maritime districts, this growth can be unplanned, leading to high-density industrial-residential zones that lack adequate green cover. The environmental cost includes the loss of coastal biodiversity and increased energy consumption for cooling. As Thoothukudi expands, it creates unique "micro-environments" that necessitate advanced geospatial monitoring to track these sprawling environmental footprints.

1.2.2 Impact of Urbanization on Local Microclimates

Urbanization alters the physical properties of the Earth's surface, specifically its albedo and thermal emissivity. In Thoothukudi, the removal of native vegetation for industrial sheds and port expansion reduces evapotranspiration—the natural cooling process. Consequently, surplus solar energy is stored in concrete structures and released at night, significantly elevating local temperatures. These microclimatic shifts lead to "hotspots" within the city,

particularly near the thermal power plants and industrial estates. Understanding these shifts is vital for protecting the district's vulnerable labor force and ensuring that the coastal infrastructure can withstand the increasing thermal demands of a warming localized climate.

1.3 THE CONCEPT OF URBAN HEAT ISLAND

The Urban Heat Island (UHI) effect is a phenomenon where urban or industrial regions experience significantly higher temperatures than their surrounding rural counterparts. This temperature disparity is primarily driven by the replacement of natural vegetation with anthropogenic surfaces that absorb and retain heat. In the context of this research, understanding the UHI effect is critical for assessing the environmental health of Thoothukudi, where rapid industrialization and coastal dynamics converge.

1.3.1 Surface vs. Atmospheric Heat Islands

Urban Heat Islands are categorized into Surface and Atmospheric types. Surface UHI (SUHI) refers to the relative warmth of urban surfaces compared to rural landscapes and is most intense during the day. This is the primary metric measured by the Landsat-8 thermal sensors used in this study. In contrast, Atmospheric UHI refers to the

warming of the air layer between buildings. In a coastal environment like Thoothukudi, the interaction between the sea breeze and surface heat creates complex thermal dynamics. This research prioritizes SUHI analysis using Land Surface Temperature (LST) to provide a high-resolution map of thermal stress across the district's varying land covers.

1.3.2 Drivers of Thermal Transformation in Industrial Cities

In an industrial hub like Thoothukudi, the thermal landscape is driven by dense built-up structures, vast salt pan modifications, and industrial emissions. Dark surfaces, such as tar roads and factory roofs common in the SIPCOT industrial area, have low albedo and absorb high amounts of solar radiation. Furthermore, the semi-arid nature of the district means a lack of "blue-green" infrastructure prevents natural cooling. Anthropogenic heat from heavy machinery, port operations, and vehicles further compounds the problem. This "Thermal Transformation" is a progressive degradation where every new industrial project contributes to a cumulative increase in the district's average surface temperature.

1.4 ROLE OF REMOTE SENSING AND GEOSPATIAL TECHNOLOGY

The integration of Remote Sensing (RS) and Geographic Information Systems (GIS) has revolutionized environmental

monitoring by providing high-resolution, multi-temporal data that is essential for tracking complex phenomena like the Urban Heat Island (UHI) effect. In a district as spatially diverse as Thoothukudi, these technologies allow for a comprehensive assessment of how industrial growth translates into thermal degradation.

1.4.1 Advantages of Landsat-8 Thermal Infrared Sensors

Landsat-8 offers a unique advantage for studying Thoothukudi's climate through its Thermal Infrared Sensor (TIRS). With a 30-meter spatial resolution, it is ideal for identifying thermal variations between the city center, the port area, and the outlying agricultural zones. Unlike ground-based weather stations in the district which provide only point-specific data, Landsat-8 provides a continuous "wall-to-wall" thermal map. This allows the research to correlate surface temperature directly with Thoothukudi's specific land-use types—such as industrial zones versus salt pans—with high geometric precision and temporal consistency from 2016 to 2025.

1.4.2 Google Earth Engine for Spatio-Temporal Analysis

Processing a decade of satellite data for the entire Thoothukudi district requires immense computing power. Google Earth Engine (GEE) overcomes this by providing a cloud-based platform for planetary-scale analysis. GEE allows for the rapid execution of scripts to mask the persistent coastal clouds and aggregate annual composites across thousands of images. By using GEE, this study

ensures high computational efficiency. The platform's ability to handle multi-temporal datasets makes it the premier tool for tracking the long-term Urban Thermal Field Variance Index (UTFVI) and identifying decadal trends in Thoothukudi's warming without the need for manual data downloads.

1.5 PROBLEM STATEMENT

Despite its rapid evolution as a critical industrial and maritime gateway, the region of Thoothukudi lacks a structured, decadal assessment of its thermal environment. Between 2016 and 2025, the city has witnessed significant land-use alterations, yet the resulting intensity of the Urban Heat Island (UHI) effect remains largely unquantified. Current environmental monitoring fails to integrate the complex relationship between surface temperature and ecological health indices. Without a clear understanding of where "thermal hotspots" are forming, urban planners cannot effectively address the escalating risks of heat stress on the workforce or the surrounding agricultural landscape. This research addresses this gap by creating a geospatial framework to decode the thermal transformation and identify high-risk zones that require immediate intervention.

1.6 NEED FOR THE STUDY

As global temperatures rise, the "local warming" caused by

urbanization acts as a force multiplier, making cities significantly more dangerous during heatwaves. There is an urgent need to move beyond simple temperature mapping toward a holistic Ecological Risk Assessment. This study is vital because it utilizes a unique 10-year dataset to provide a long-term perspective on environmental degradation. By calculating the Heat Health Risk Index (HHRI) and Crop Water Stress Index (CWSI), the research provides actionable data that can protect public health and optimize water resource management for local agriculture. In an era of climate change, this geospatial evidence is essential for transitioning from reactive disaster management to proactive, sustainable urban cooling.

1.7 AIM AND OBJECTIVES

The primary aim of this research is to develop a comprehensive geospatial framework using Google Earth Engine (GEE) to analyze the spatio-temporal thermal transformation of Thoothukudi and evaluate its subsequent ecological and health impacts from 2016 to 2025.

1.7.1 Objectives

- To generate Landsat-8 TIRS and OLI sensors annual cloud-free composites for the study area via Google Earth Engine.
- To quantify thermal indices namely LST and biophysical indices namely NDVI, NDBI, FVC to track land-cover changes.
- To categorize the ecological quality of the urban environment

based on heat island dynamics namely Urban Heat Island (UHI) and the Urban Thermal Field Variance Index (UTFVI) by random forest technique.

- To Model Ecological Risk through Heat Health Risk Index (HHRI) and Crop Water Stress Index (CWSI).
- To Perform statistical validation of the derived models and suggest data-driven policies and urban resilience.

CHAPTER 2

REVIEW OF LITERATURE

2.1 GENERAL

The literature review provides the theoretical framework necessary to understand the complex interactions between urbanization and microclimatic shifts. As global temperatures rise, the Urban Heat Island (UHI) effect has transitioned from a meteorological curiosity to a critical environmental challenge, necessitating rigorous academic inquiry. This chapter synthesizes previous research concerning the retrieval of Land Surface Temperature (LST) and the application of multispectral indices such as NDVI, NDBI, and NDWI for environmental monitoring. By evaluating the evolution of geospatial technologies—from traditional image processing to modern cloud-based analytics via Google Earth Engine—this review establishes the state of the art in thermal mapping. Furthermore, it examines the integration of ecological models like HHRI and CWSI, ultimately identifying the research gaps that this decadal study of Thoothukudi aims to bridge.

2.2 STUDIES ON URBAN HEAT ISLAND

Britta Stumpe et al., (2026) Study investigates In the context of global urbanization and increasingly frequent heatwaves, urban green spaces (UGS) play a crucial role in mitigating urban heat islands

(UHI). However, the variability and reliability of these cooling services is often overlooked. This study addresses the temporal and inter-site variability of cooling effectiveness across different UGS types in Berlin, Germany. We analyzed inner (land surface temperature, LST) and outer (park cooling distance, PCD; park cooling intensity, PCI) cooling parameters for 228 parks, 512 allotments, and 143 cemeteries using six cloud-free Landsat 8 OLI scenes from hot summer days ($T_{\max} > 25^{\circ}\text{C}$). While LST showed low variabilities between sites, PCD and PCI fluctuated more, especially in allotments. Only 9-16% of the UGSs exhibited stable cooling distances over time, with parks achieving the largest PCD and cemeteries providing the most intense and reliable cooling. Allotments were the least effective and most variable. To investigate drivers of temporal PCD variability, we conducted regression analyses using vegetation indices, spatial metrics, and urban structure. However, the models explained little of the variance (R^2 up to 0.3), indicating complex external influences not fully captured by the available data. To analyze the multifaceted role of UGS in UHI mitigation, we propose the novel concept of "Cooling Relevance" which integrates intensity, spatial extent, reliability, and the number of potential beneficiaries. Parks emerged as the most relevant UGS type under this framework. Our findings highlight the need to incorporate reliability into urban greening strategies to support equitable and climate-resilient urban planning.

Shuyang Zhang et al., (2026) Research investigates the intersection of climate change, extreme weather (EW) events like heatwaves and cold snaps are becoming more frequent, challenging urban buildings and energy systems. Urban heat island (UHI) effects- where city centers are significantly warmer than suburbs at night- further impact heating and cooling demands of urban buildings. However, there is still a lack of systematic empirical studies linking meteorological data to building energy use, especially concerning the compound effects of UHI and extreme weather on urban building energy consumption and load during both winter and summer. To address future complex climate conditions, we propose using localized weather data (LWD) that fully accounts for both background EW and UHI effects. Driven by suburban meteorological observations and high-resolution land cover data, the data are generated using the UWG urban canopy model and the UMEP tool on the QGIS platform to capture realistic local weather conditions around buildings. It can be directly input into urban building energy model (UBEM) for the corresponding local climate zones to simulate building energy use. Our study shows that LWD better captures seasonal building energy use and the effects of external and internal factors. Compared to suburban weather station data, the accuracy improves by 29.6% in summer and 36.6% in winter during the typical year, and by 35.1% and 30.1% during the extreme weather year, respectively. Local air temperature (T_a) has the greatest impact on actual energy use, followed by solar radiation (Rad)- especially during summer

heatwaves, when R_{ad} may exceed T_a in influence. Internal disturbances have a greater impact in summer, but their influence lessens during extreme weather due to stronger external climatic effects. This method supports refined assessment and control of UBEM across climates and seasons, helping manage energy peaks during heatwaves and prevent overheating in winter, ultimately aiding real-weather-based energy system optimization and urban design.

Elizabeth Berg et al., (2026) Observes due to differences in energy balance, urban areas typically experience greater temperatures than surrounding rural areas, a phenomenon known as the urban heat island (UHI) effect. This urban-rural temperature difference can vary greatly, and the magnitude of the UHI depends on the time of day, season, and weather conditions. This study used 12 years of observational records from an extensive sensor network in Madison, Wisconsin, to explore patterns in short and long-term variability of the local UHI, as well as the meteorological drivers of this variability. We identified a statistically significant year-round inverse correlation between wind speed and UHI magnitude. For other meteorological variables, correlation with the UHI varied seasonally, with the strongest dependence on temperature occurring during the winter and the strongest dependence on precipitation occurring during summer and fall. We also observed a greater frequency of inverted heat islands days when urban temperatures dropped below rural

temperatures - than previously documented in Madison or climatically similar cities. Additionally, we identified marked differences between spring and fall conditions that point to the role of prior seasons in shaping the UHI. Overall, the trends documented here have the potential to inform the development and evaluation of heat mitigation efforts in the Madison area. More broadly, the degree of variability observed in this study relative to prior analyses highlights the benefits that long-term, year-round monitoring can provide to urban climate planning throughout the world.

Vannesah K. Kporha et al., (2026) Assessed the utility of MODIS daily and 8-day composite data in modelling land Surface Temperature (LST) is a critical metric for understanding surface-atmosphere interactions and serves as a key variable in analyzing the Urban Heat Island (UHI) effect, a phenomenon where urban areas exhibit elevated temperatures compared to surrounding rural regions. UHI effects, driven by urbanization and surface thermal properties, significantly impact urban climates, particularly under a changing climate. This study evaluates the utility of MODIS daily and 8-day composite data in modeling summer LST and UHI effects across 137 cities in continental France over a 10-year period. Using Random Forest machine learning model and SHAP analysis, we assessed the role of structural, temporal, and meteorological variables in predicting LST and UHI dynamics for both daytime and nighttime. The

results reveal that MODIS daily data achieves higher model accuracy ($R^2 = 0.85$ for daytime, $R^2 = 0.85$ for nighttime) compared to 8-day composites ($R^2 = 0.75$ for daytime, $R^2 = 0.7$ for nighttime) when meteorological inputs are included. However, in scenarios without weather data, the MODIS 8-day data outperformed daily data for daytime LST prediction ($R^2 = 0.57$ vs. $R^2 = 0.48$), emphasizing its potential for regions with limited meteorological coverage. Month emerged as a key predictor across all models, serving as a proxy for seasonal temperature variability. Diurnal analyses revealed stronger daytime UHI prediction accuracy whereas nighttime models showed lower accuracy, particularly for 8-day composites. Overall, this study clarifies when coarser-temporal satellite products can reliably substitute daily observations and identifies the dominant drivers of urban heat, providing practical guidance for large-scale UHI monitoring and heat-resilient urban planning in regions with limited ground-based meteorological data.

Noushin Mokhtarzadeh et al., (2026) Investigated the day-to-day relationship between the Urban Heat Island (UHI) effect and precipitation in Houston, Texas, using a data-driven, multi-layer spatiotemporal framework. By integrating satellite observations, reanalysis datasets, and wind-field information with machine learning based reconstruction of missing thermal data, the approach overcomes observational gaps and identifies atmospheric conditions favorable for UHI-precipitation interactions. Results show that

approximately 290 out of 800 critical days exhibit strong alignment between UHI intensity, prevailing wind direction, and rainfall occurrence, confirming that UHI-induced precipitation enhancement is highly conditional and direction-dependent. Thermo-hygrometric clustering further reveals four atmospheric regimes linking temperature, humidity, and UHI intensity to rainfall patterns. Notably, nearly 48% of all events occur within a warm-humid regime characterized by mean temperature of 28.3°C, mean relative humidity of 83.6%, and mean UHI intensity of 5.8°C, conditions that promote convective uplift and moisture convergence. The findings demonstrate that static or mean-based analyses obscure transient feedback between urban heating and atmospheric dynamics. The proposed framework highlights the necessity of combining machine learning, downwind diagnostics, and high-resolution temporal coupling to better understand complex UHI-rainfall interactions, offering actionable insights for urban planning, hazard mitigation, and climate-resilient infrastructure design.

Shalini Kumari et al., (2026) conducted a longitudinal study on the impact of rapid urbanization on thermal stress and ecosystem degradation are being accelerated by rapid urbanization in ecologically sensitive mountain valleys. Nevertheless, there are still few integrated multi-decadal evaluations that connect coupled thermal-ecological processes with LULC transitions in Himalayan

ecosystems such as Doon Valley. This paper fills this gap with a new multi-index model that examines 31 years of Landsat data (1993-2024) to measure urbanization driven LULC changes, LST warming, and UHI impacts. Based on integrated spectral indices (NDVI, SAVI, NDBI, and BSI), thermal measures (LST, UTFVI, and EEI), we find: (i) urban growth of 133% mainly due to agricultural conversion; (ii) 66% agricultural loss; (iii) average LST growth of 9.17°C accelerating Constructed/bare surfaces are 34-40°C whereas vegetated are colder (26-32°C). The rigorous multi-sensor validation (OA 93-97% and Kappa 0.91-0.96) and multi-index integration give the world the new insights of spatial heterogeneity, showing the loss of impervious surfaces and blue-green infrastructure as the thermal drivers, as equal to single-factors analysis. The spatial hotspot maps and ecological vulnerability indices in this investigation give practical advice to municipal planners at the ward scale, including the need to protect riparian corridors/wetlands, increase city vegetation in the form of parks, street trees, and green roofs, rehabilitate water bodies and natural drainage, encourage compact development with mixed uses, and climate-adaptive land-use strategies with a primary focus on agricultural protection. It can be also supported by SDG 11 (sustainable cities), SDG 13 (climate action). and SDG 15 (life on land) because the replicable framework can provide evidence-based approaches to sustainable development under the conditions of thermal-hydrological-ecological challenges of rapidly urbanizing mountainous areas.

Anmol Gaur et al., (2026) conducted a comprehensive review on the evolving role of Machine Learning (ML) in mitigating and analyzing the Urban Heat Island (UHI) effect is a phenomenon in which urban areas experience significantly higher temperatures than their surrounding regions due to human activities and the built environment. This poses challenges to urban systems, living conditions, and exacerbate air pollution in urban areas. Over the years, the use of Machine Learning (ML) to study the UHI effect has increased, driven by advances in computational power, high predictive accuracy, and the growing availability of large-scale, environmental datasets. This study aims to comprehensively review the application and role of ML in addressing the UHI effect. The key findings of this review show that the application of ML can be categorized into: land surface temperature (LST) retrieval and prediction, land use land cover (LULC) classification and change detection, UHI intensity and pattern modelling, and mitigation strategy assessment. The most common application is the derivation and prediction of LST in conjunction with physical and climatic variables such as building energy consumption, air temperature, human thermal comfort, and wind movement. Our findings further show that ML is effectively utilized in multi-temporal investigations and predictive modelling, assessing energy demand surges, and validating the efficacy of cooling interventions. We also outline the

key challenges faced in implementing ML. in this realm, including aerial data accessibility, data pre-processing complexities, modelling sophistication or UHI, and lack of ground truth data for validation. This review sets the stage for future use of ML, including emerging Artificial Intelligence (AI) driven methods that could change how we understand and mitigate UHI effects.

Saskia Rupp et al., (2026) Investigated the latest Conference of the Parties (COP) demonstrated that development and 'Climate change' are inextricably intertwined, making them the primary fields to be tackled in addressing rational global tendencies. Since the beginning of urbanization, asphalt, and concrete have supplanted natural vegetation, which has led to an increase in energy demand. This increase has accelerated the urban heat island (UHI) effect and paved the way for a rise in building energy consumption. In addition to that, the increased impervious surfaces capture more solar radiation, increasing Land Surface Temperature (LST). Increasing Greenspace, where vegetation changes have been monitored with the help of the Normalised Difference Vegetation Index (NDVI) since the early 1980s, has been suggested as a strategy for urban areas in response to increasing urbanization and forecasts of more frequent heatwaves under a climate change scenario. The present study focuses on using remotely sensed data of MODIS to evaluate the relationship between, Urban Greenspace and UHI for the city of

Gandhinagar, Gujarat. For the detection of relation as well as change, Land Surface Temperature ($^{\circ}\text{C}$) and Precipitation rate (mm/day) were incorporated. This study compared UHI and NDVI for an annual average for the years 2003 and 2018. In the entire study period, a considerable change in land use was observed with a significant increase in build-up area. Correlation among the variables depicted a significant negative relation between NDVI ($r=-0.76$) and LST and Precipitation ($r=-0.38$). The linear

Wenjing Yang et al., (2025) Addressed the Urban trees play a pivotal role in urban ecosystems, effectively reducing urban surface temperature by transpiration induced evaporative cooling. However, accurately estimating tree transpiration (E_c) in urban environments is challenging due to a lack of robust methods and the inherent complexities of soil moisture monitoring in root zones, especially during dry periods. In this study, a simplified mechanistic tree transpiration model (BTA, Buckley, Turnbull & Adams) was improved by a new species-dependent normalized canopy temperature water stress index (NCTI), based on within-canopy air temperature (T_c) and reference temperature of an irrigated park (T_1). The performances of the new models (BTA-T) were tested using sap flow observations from two species (White Cedar: *Melia azedarach*; Grey Box: *Eucalyptus microcarpa*), with a total of seven trees in the City of Mitcham, South Australia. Results show that the BTA- T_e models reduced the

overestimation of the BTA model noticeably during dry periods. Overall, the Nash-Sutcliffe efficiency (NSE) increase was 10% for both White Cedar and Grey Box trees compared to the BTA model. Additionally, NCTI effectively captured continuous seasonal tree water stress patterns, determined by the morning air warming rate for White Cedar trees and by the afternoon average temperature for the Grey Box trees. The NCTI also revealed the benefit of a Water Sensitive Urban Design device in mitigating street tree water stress during the early dry season. Given its simple data requirements, the BTA-Te model can be easily applied to other locations, species, or climates, offering a practical approach for monitoring urban tree water use and assessing green infrastructure performance under variable climate conditions.

Tsegaye Ayele Tadese et al., (2025) This study investigated the Urban Heat Island (UHI) effect in Addis Ababa, Ethiopia, caused by rapid urbanization and changes in land use and land cover (LULC). The problem of increasing land surface temperatures (LST) in urban areas is linked to the expansion of impervious surfaces like asphalt and buildings, which exacerbate heat retention and impact the environment and public health. Using Landsat 8 satellite imagery, the study analyzed LST variations through remote sensing techniques, focusing on the Normalized Difference Vegetation Index (NDVI) and the Normalized Difference Built-Up Index (NDBI). Regression models,

including Ordinary Least Squares (OLS) and Geographically Weighted Regression (GWR), were applied to assess the spatial relationship between LST and LULC changes. The results revealed that vegetated areas had a strong negative correlation ($r = -0.8293$) with LST, indicating a cooling effect. In contrast, built-up areas showed a positive correlation ($r = 0.4491$) with LST, contributing to higher temperatures. LST values in the city ranged from 10°C to 44°C , with the highest temperatures recorded in urbanized areas and the lowest in vegetated zones. These findings emphasize the importance of expanding green spaces to mitigate the UHI effect. The study's results are vital for guiding urban planning policies aimed at reducing heat-related risks in rapidly urbanizing cities, helping achieve SDG 11 (Sustainable Cities and Communities) and SDG 13 (Climate Action). The integration of green infrastructure can play a critical role in reducing the adverse impacts of urban heat islands.

Hoang Thi Hang et al., (2025) Investigated the rapid urbanization of Delhi has significantly transformed its Land Use and Land Cover (LULC), leading to profound alterations in Land Surface Temperature (LST) and exacerbating environmental challenges. Therefore, this study integrates multi-temporal Landsat imagery, advanced machine learning techniques, and landscape fragmentation indices to assess the impact of urban expansion on LST from 2001 to 2021. LULC classification was conducted using the Random Forest algorithm,

achieving high classification accuracy. Landscape metrics such as Number of Patches (NP), Largest Patch Index (LPI), Mean Patch Area (MPA), Patch Density (PD), Landscape Shape Index (LSI), and Fractal Dimension (FD) were computed to quantify urbanization trends. The mono-window algorithm was used for LST retrieval, and a spatial regression framework incorporating Spearman's correlation and multivariate regression models was applied to establish statistical relationships between urbanization metrics and temperature variations. Results indicate a 50 % increase in built-up area, accompanied by an 81 % reduction in open land and a 15 % decline in cropland over two decades. LST exhibited a substantial rise, with the mean temperature in built-up regions increasing from 36.58 °C in 2001 to 41.81 °C in 2021, and peak temperatures reaching 51.60 °C, highlighting the intensification of the urban heat island (UHI) effect. Regression analysis revealed a strong negative correlation between LPI and LST ($R^2 = 0.99$), indicating that fragmented urban structures contribute to higher temperatures. Conversely, LSI and FD showed positive correlations with LST, confirming that irregular and dispersed urban growth patterns exacerbate thermal stress. These findings underscore the need for sustainable urban planning, emphasizing compact city development, green infrastructure, and water body conservation to mitigate UHI effects.

M.Kirubakaran et al., (2026) Addressed In hydrogeological

stations, missing data is a frequent issue, and filling in the missing value can be difficult. The current study presents a novel and automated method for assessing the long-term groundwater level (GWL) trend in arid and semi-arid regions using a hybrid imputation framework that effectively fills missing GWL data by preserving both spatial and temporal relationships, and comprehensive spatio-temporal trend analysis, which includes Mann-Kendall trend analysis and GIS-based spatial assessment. The entire research process was carried out in a Python environment, which approached GWL studies fully automated, scalable, more effective, and guaranteed reproducibility of the research work. The integrated approach adopted in the current study provides methodological strength and scientific novelty by combining the best-suited imputation with spatial trend analysis to better understand GWL dynamics in arid and semi-arid regions. The study used 21 years of GWL data from 42 monitoring wells for the Thoothukudi district obtained from the State Data Centre of Tamil Nadu. The temporal trends were analysed annually and seasonally using the Mann-Kendall (MK) test to detect monotonic patterns, while spatial trends were assessed using a geographical information system (GIS). The proposed hybrid imputation model effectively handled missing data while preserving spatio-temporal dependencies, achieving high accuracy with a mean RMSE value of 0.863 and a mean R value of 0.793. The MK trend analysis revealed statistically significant declining trends ($p < 0.05$) in several wells, predominantly in

the southwestern, southern, eastern, and parts of the central region of the study area. The MK trend analysis for each season identified critical stress periods during winter and summer seasons, with z-score declines exceeding -2.5 in some wells. The present study's findings emphasise the urgent need for sustainable groundwater management, particularly in the eastern, southwestern, and southern zones with pronounced declining levels.

Arunima Nandy et al., (2025) Examined land surface temperature (LST) and vegetation play a very significant role in influencing soil moisture and its interaction with the crop. Crop Water Stress Index (CWSI) is one of the most applicable indices to measure the level of water stress to crops that can be quantified using satellite imagery. The upper Brahmaputra river valley of Assam is a rapidly growing region experiencing extensive urban expansion, which necessitates the assessment of changes in spatio-temporal LST and vegetation dynamics for sustainable land management. MODIS images were applied in this study to assess spatio-temporal LST, vegetation and Crop Water Stress (CWS). Normalized Difference Vegetation Index (NDVI) was adopted to assess the spatio-temporal vegetation dynamics. A correlation study was conducted to understand the relationship between LST, NDVI and CWSI. It was observed that there was a strong negative correlation between LST and NDVI, whereas a strong positive correlation was found between

LST and CWSI. Hotspot areas characterized by high temperature, low vegetation and high crop water stress were delineated in the ArcGIS platform. Between 2001 and 2021, all LST zones showed an increase in both maximum and minimum temperatures. Contour tilling, mulching and shade nets may effectively enhance soil moisture retention, fertility and microclimatic conditions in these hotspot areas

Kaushik Mandvikar et al., (2024) Explored the latest Conference of the Parties (COP) demonstrated that development and 'Climate change' are inextricably intertwined, making them the primary fields to be tackled in addressing rational global tendencies. Since the beginning of urbanization, asphalt, and concrete have supplanted natural vegetation, which has led to an increase in energy demand. This increase has accelerated the urban heat island (UHI) effect and paved the way for a rise in building energy consumption. In addition to that, the increased impervious surfaces capture more solar radiation, increasing Land Surface Temperature (LST). Increasing Greenspace, where vegetation changes have been monitored with the help of the Normalised Difference Vegetation Index (NDVI) since the early 1980s, has been suggested as a strategy for urban areas in response to increasing urbanization and forecasts of more frequent heatwaves under a climate change scenario. The present study focuses on using remotely sensed data of MODIS to evaluate the relationship between, Urban Greenspace and UHI for the city of

Gandhinagar, Gujarat. For the detection of relation as well as change, Land Surface Temperature ($^{\circ}\text{C}$) and Precipitation rate (mm/day) were incorporated. This study compared UHI and NDVI for an annual average for the years 2003 and 2018. In the entire study period, a considerable change in land use was observed with a significant increase in build-up area. Correlation among the variables depicted a significant negative relation between NDVI ($r=-0.76$) and LST and Precipitation ($r=-0.38$). The linear

Anitha V et al., (2024) Investigated the present study involved analyzing the impact of Normalized Difference Vegetation Index (NDVI), a measure of vegetation density, on Land Surface Temperature (LST) and its role in the formation of Surface Urban Heat Island (SUHI) effect in Alappuzha district, Kerala. Imagery from Landsat TM, Landsat OLI and TIRS for the years 1988 and 2020 was utilized to evaluate the correlation between NDVI and SUHI. The land area corresponding to each NDVI category were quantified. The results revealed that NDVI of the study area declined from 1 in 1988 to 0.56 in 2020, resulting in an increase in maximum and mean LST. LST increased from 32.98°C in 1988 to 37.44°C in 2020. LST Mean which was 25.66°C in 1988 increased to 26.88°C in 2020. The decrease in NDVI led to a rise in the high SUHI classes such as Middle, Strong and Stronger by 128.62%, 47.66% and 47.66% respectively. A negative correlation was observed between NDVI and LST, with Pearson

coefficients of 0.4874 for 1988 and 0.4097 for 2020, indicating that NDVI might be utilized as an important indicator to analyze LST and SUHI effects.

Timsi Modi et al., (2024) Emphasized the latest Conference of the Parties (COP) demonstrated that development and 'Climate change' are inextricably intertwined, making them the primary fields to be tackled in addressing rational global tendencies. Since the beginning of urbanization, asphalt, and concrete have supplanted natural vegetation, which has led to an increase in energy demand. This increase has accelerated the urban heat island (UHI) effect and paved the way for a rise in building energy consumption. In addition to that, the increased impervious surfaces capture more solar radiation, increasing Land Surface Temperature (LST). Increasing Greenspace, where vegetation changes have been monitored with the help of the Normalised Difference Vegetation Index (NDVI) since the early 1980s, has been suggested as a strategy for urban areas in response to increasing urbanization and forecasts of more frequent heatwaves under a climate change scenario. The present study focuses on using remotely sensed data of MODIS to evaluate the relationship between, Urban Greenspace and UHI for the city of Gandhinagar, Gujarat. For the detection of relation as well as change, Land Surface Temperature (°C) and Precipitation rate (mm/day) were incorporated. This study compared UHI and NDVI for an annual

average for the years 2003 and 2018. In the entire study period, a considerable change in land use was observed with a significant increase in build-up area. Correlation among the variables depicted a significant negative relation between NDVI ($r=-0.76$) and LST and Precipitation ($r=-0.38$). The linear.

Sang yong chung et al., (2023) Investigated the aspects of ecological issues arising from rapid urbanization are investigated in this paper using Land Surface Temperature (LST) in terms of radiation budgets studied by heat conduct balance using a geospatial approach. It uses the data from USGS Landsat ETM+ 4-5 and Landsat-8 OLI imagery acquired on February 6, 1988 and October 14, 2021. The geospatial technique examines the effects of changes in Land Use/Land Cover (LULC) in terms of NDVI classification on the distribution of surface temperature. The remote sensing approach identifies changes in land use, their influence on surface temperature, and variations in average LST caused by these hotspots. This study examines the distribution of LST in Kancheepuram district, Tamil Nadu, India. The LSTs ranged from 14 deg * C to 31 deg * C in 1985, from 25 deg * C to 39 deg * C in 2005, and from 31°C to 47 deg * C in 2021. The results show a 152% increase in the mean LST in 4 decades. The LST high values were observed in urban and industrial regions with buildings, impervious sidewalks, and sparse vegetation. The intensity at hotspots was highest in urbanized and sparsely

vegetated regions. The low Urban Heat Island (UHI) effect was measured in vegetated areas. Industrial areas and parking lots were commonly classified as high-intensity hotspots. The elevated temperatures are attributable to the materials used in the construction at these locations. Further research is needed to better understand the reflectance of thermal radiations by certain materials used in urban regions and how to minimize temperatures in such locations.

Masilamani Palanisamy et al., (2023) Highlighted In the history of humankind, urbanization is one of the vibrant changing geographical phenomena. Almost 55% of the world's population lives in rapidly urban and surrounding areas. Urban heat island represents a significant environmental problem most urban centres face. Extreme heat in urban areas reflects substantial risks to the growing urban population. The increased surface temperature is due to the thermodynamic characteristics of the built-up in urban. The impact of land use/land cover (LULC) dynamics is closely related to urban heat islands (UHI). This study aims to identify the urban heat zones by utilising indicators such as the LULC, NDBI, LST and NDVI for Tiruppur Corporation. The study area Tiruppur is known as the knitwear capital of India, attracting thousands of migrants from all over the country and leading to a continuous accumulation of people. So, most of the migrant people settled in the Tiruppur belt. The

present study analysed the spatio-temporal land-use patterns, NDBI, LST, NDVI and UTFVI for Tiruppur Corporation using Landsat and sentinel data sets for 1991, 2001, 2011 and 2021. The LULC were prepared by utilising the supervised classification algorithm of Support Vector Machine (SVM). On the other hand, the advanced technique of Google Earth Engine is utilised to map the LST and NDVI of the study area for the same year. All the results of four indicators, i.e., LULC, NDBI, LST, NDVI and UTFVI, are analysed with the temporal changes by segmenting them with six rings of each 2km. This will help to identify the changes and increase of urban heat from the core to the periphery areas of the Tiruppur Corporation. Thus, this study aids in improving future urban planning, including implementing green city technologies.

Tageui Hong et al., (2023) Investigated the urban and meteorological factors strongly affect street-level urban heat islands (UHIs), but few studies have considered their interactions under varying weather conditions. This study investigated the relationship between urban and meteorological factors and street-level UHI intensity in Seoul during daytime and nighttime in summer. UHI intensity was calculated from urban air temperatures measured by 568 street-level sensors. Random forest regression models and Shapley Additive exPlanations (SHAP) method were used to quantitatively analyze nonlinear relationships and interaction effects

of the predictors. The results indicated that meteorological variables, particularly regional air temperature, significantly influenced UHI intensity during both daytime and nighttime. Furthermore, urban factors such as building coverage ratio and pervious ratio became more important during nighttime. Both meteorological and urban variables indicated nonlinear relationships with UHI intensity, with some showing threshold effects. Compared to total effects, main effects were significantly smaller in magnitude and range due to high parameter interactions among variables. For most variables, the sum of interaction effects outweighed main effects. In particular, notable interaction effects were observed within each meteorological and urban category. These results pinpoint that effects of urban variables are important individually and in combination with other variables. The findings highlight the importance of designing effective mitigation strategies that account for both nonlinear relationships of individual factors influencing UHI and interactive influences of multiple factors.

Priyanka Rao et al., (2023) Conducted an long term temporal analysis of imola city. urbanization strongly correlates with land use land cover (LULC) dynamics, which further links to changes in land surface temperature (LST) & urban heat island (UHI) intensity. Each LULC type influences UHI differently with changing climate, therefore knowing this impact & connection is critical. To understand such

relations, long temporal studies using remote sensing data play promising role by analysing the trend with continuity over vast area. Therefore, this study is aimed at machine learning centred spatio-temporal analysis of LST and land use indices to identify their intra-urban interaction during 1991–2021 (summer) in Imola city (specifically representing small urban environment) using Landsat- 5/8 imageries. It was found that LST in 2021 increased by 38.36% from 1991, whereas average Normalised Difference Built-up Index (NDBI) increased by 43.75%, associating with increased thermal stress area evaluated using ecological evaluation index. Major LULC transformations included green area into agricultural arable-land and built-up. Finally, the modelled output shows that built-up & vegetation index have strongly impacted LST. This study, help to understand the relative impact of land-use dynamics on LST at intra-urban level specifically with respect to the small urban settings. Further assisting in designing and regenerating urban contexts with stable configuration, considering sustainability and liveable climate, for benefit of health of public and fragile population in particular.

Mohammed Faizan (2020) Explores the broader environmental consequences of rapid urbanization has led to many important problems, such as rising levels of pollution, rapid climate changes and rising urban temperatures, that is the formation of Urban Heat Islands (UHI) As the population density increases, most land areas are

being transformed into cities and they are growing very rapidly. The cities are getting hotter day by day because of the UHI effect. Remote sensing and GIS have been found to be an effective tool for analysing this phenomenon. In urban environmental issues, Urban Heat Island plays an important role. This present study investigates the spatial and temporal pattern and changes of urban heat island (UHI) in Chennai city Landsat 7 and 8 Imageries of 2000, 2010 and 2020, whose spatial resolution sufficient for some significant environmental parameters to be calculated. Land surface temperature (LST) values were extracted from Landsat 7 ETM+ (2000,2010) and Landsat 8 OLI (2020). A Normalized Difference Vegetation Index (NDVI), Normalized Difference Built-up Index (NDBI) maps was extracted from the same image. The thermal data were corrected and converted to top of the atmosphere radiance values and further to temperature in degrees Celsius from Kelvin scale using ArcGIS. Four land use and land cover (LULC) types (Vegetation, Barren land, built-up area and water) were classified from the Landsat images by supervised classification techniques using maximum likelihood classification (MLC) algorithm. LULC, NDVI, NDBI, LST maps have been prepared using ArcGIS 10.8. Result indicate that during the investigation period (2000-2020), high temperatures were typically experienced mostly in concentrated built-up areas with lower concentrated temperatures in vegetation areas. From the results, it is clear that the values of land surface temperature had a positive correlation with NDBI and a negative correlation with NDVI This

research can be a data source for urban planners, designers, scientists, engineers, demographers and other social scientists to improve the quality of the environment by designing green lands, changing the surface cover and developing guidelines for urban design, as well as making decisions that will improve sustainability methods in the environment.

CHAPTER 3

STUDY AREA AND ITS DESCRIPTION

3.1 GENERAL

This chapter provides a comprehensive description of the study area, focusing on its geographical, environmental, climatic, and socio-economic characteristics, which are essential for understanding spatial patterns and interpreting geospatial analysis results. The

study area selected for this research is Thoothukudi district, located in the southern part of Tamil Nadu, India. Thoothukudi is a coastal district that holds significant importance due to its unique combination of natural resources, industrial activities, and environmental conditions. The district is widely recognized for its port, salt production, and fisheries, along with its expanding industrial base. From a geospatial perspective, the region offers diverse land use and land cover features, including agricultural lands, coastal zones, urban settlements, and industrial areas. These variations make it highly suitable for remote sensing and GIS-based analysis. Moreover, the district experiences notable environmental changes influenced by both natural processes and anthropogenic activities, such as urbanization and industrialization. Studying this area helps in understanding the interaction between land, water, and human activities. Therefore, a detailed understanding of the study area is crucial, as it forms the foundation for analyzing satellite data and deriving meaningful conclusions in subsequent chapters of this research.

Thoothukudi, historically known as Tuticorin and often referred to as the "Pearl City," is a strategically significant coastal district situated in the southeastern extremity of Tamil Nadu, India. Geographically, the study area is positioned between 8°19'N and 9°20'N latitude and 77°40'E and 78°10'E longitude. It is bordered by the Gulf of Mannar to the east, providing a vital maritime gateway

through its major all-weather port. The district's landscape is a complex mosaic of urban centers, sprawling industrial estates such as SIPCOT, extensive salt pans, and sensitive coastal ecosystems. This region was specifically selected for research due to its rapid industrial transformation and the diverse nature of its land cover. The interaction between its semi-arid inland topography and the maritime influence of the Gulf of Mannar makes it a critical location for analyzing the spatio-temporal dynamics of the Urban Heat Island effect and the subsequent thermal transformation occurring between 2016 and 2025.

The presence of expansive salt pans is a particularly significant factor for this research. These areas exhibit distinct surface thermal properties (high albedo) compared to the dense urban built-up areas or vegetated zones. The high reflectivity of the salt pans interacts with incoming solar radiation in ways that contrast sharply with the heat-retaining impervious surfaces of the industrial zones. Understanding the thermal variance between these salt pans, the coastal dunes, and the concrete-heavy urban centers is fundamental to mapping the spatio-temporal dynamics of the UHI effect in the region.

The choice of the 2016–2025 study period is deliberate and provides a longitudinal perspective on the region's thermal transformation. This decadal window captures a phase of accelerated industrial and infrastructure development in Thoothukudi, marked by

significant projects such as the expansion of the Thoothukudi Port, the implementation of new industrial policies, and substantial housing growth. By analyzing satellite imagery from this specific timeline, the research is able to map the precise trajectory of land-use change and correlate it with shifts in surface temperature.

Furthermore, the availability of high-resolution remote sensing data from Landsat and Sentinel satellites during this decade allows for a statistically robust analysis, ensuring that the detected spatio-temporal dynamics reflect genuine environmental trends rather than transient anomalies.

Table 3.1 Administrative Divisions of Thoothukudi

S.NO	TALUK	AREA (sq.km)
1	Thoothukudi	377.92 km ²
2	Tiruchendur	368.39 km ²
3	Srivaikuntam	386.38km ²
4	Ettayapuram	394.61km ²

5	Vilathikulam	1094.57km ²
6	Kovilpatti	583.18km ²
7	Kayathar	401.88km ²
8	Sathankulam	306.01km ²
9	Eral	226.83km ²
10	Alwarthirunagari	201.45km ²
11	Udangudi	404.18km ²
	TOTAL	4,745.40km ²

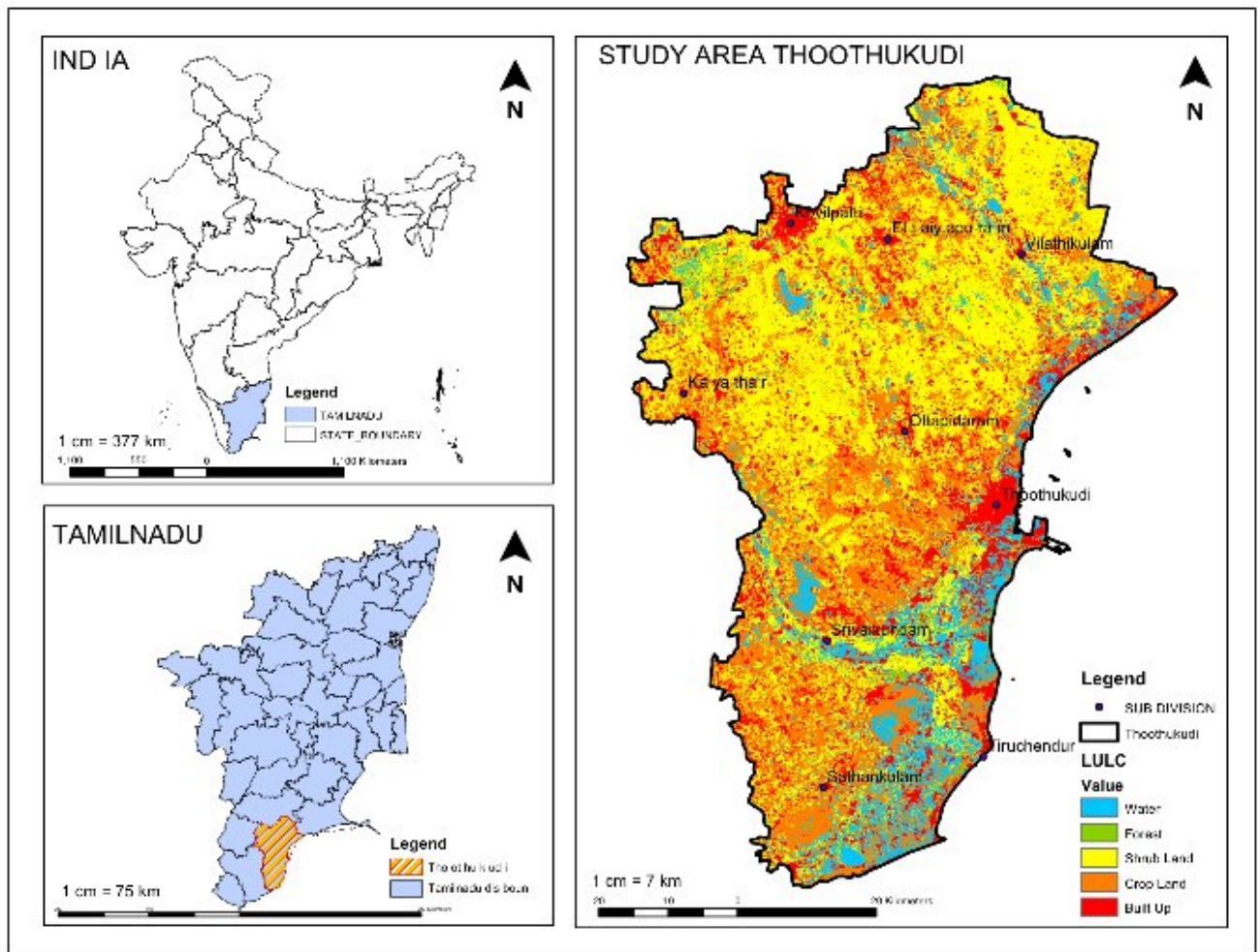


FIGURE 3.1 GEOGRAPHICAL LOCATION AND BOUNDARY MAP OF THE THOOTHUKUDI STUDY AREA

3.1.2 Climate and Topography

The district is characterized by a semi-arid tropical climate. It experiences high temperatures throughout the year, with a mean maximum temperature often exceeding 37°C during the summer months (April–June). The topography is generally flat, with a gentle slope towards the coast. Rainfall is primarily received during the North-East monsoon, making the region prone to seasonal moisture variations. These climatic conditions, combined with the low albedo of industrial materials, create a unique environment for the intensification of the Urban Heat Island (UHI) effect.

3.1.3 Socio- Economic and Industrial Profile

Thoothukudi is known as the "Pearl City" and is a major hub for thermal power generation, chemical industries, and salt production. The presence of the V.O. Chidambaranar Port has catalyzed urban sprawl and industrial growth over the last decade. The shift from traditional salt pans and agriculture to heavy industrial land use is a primary driver of the "Thermal Transformation" analyzed in this study. This rapid industrialization has transformed the district from a traditional agricultural economy into a major industrial corridor. The emergence of large-scale thermal power plants, fertilizer units, and chemical processing facilities has significantly altered the land surface characteristics. The resulting transition from pervious agricultural lands to impervious surfaces, such as concrete factory floors, asphalt roads, and sprawling storage warehouses, serves as the primary catalyst for the "Thermal Transformation" being

investigated in this study. The region's dense workforce and subsequent residential development further intensify the spatial footprint of the Urban Heat Island effect.

3.1.4 Hydrology and Environmental Sensitivity

Thoothukudi's coastal position is defined by the Gulf of Mannar, a marine biosphere reserve of significant ecological importance. The district's hydrology is primarily shaped by the Tamiraparani River and a network of minor water bodies and coastal wetlands. These water bodies play a dual role: they act as essential buffers against thermal heat gain, yet they are increasingly encroached upon by industrial expansion and salt pan conversion. The proximity to the coast provides a moderating maritime breeze; however, the inland penetration of this cooling effect is increasingly obstructed by the dense vertical growth of industrial infrastructure and residential high-rises.

3.1.5 Rationale for Study Area Selection

The selection of Thoothukudi as the primary study area is motivated by its unique status as a "living laboratory" for urban and industrial thermal expansion in a semi-arid coastal setting. Unlike inland urban centers, Thoothukudi presents a complex interaction between:

Natural Thermal Regulators: Coastal breezes and wetlands.

Anthropogenic Heat Drivers: Massive industrial clusters, rapid urban sprawl, and the systematic conversion of salt pans into land banks for

commercial use.

This juxtaposition provides an ideal environment to test hypotheses regarding the "thermal footprint" of industrialization. By analyzing the longitudinal data from 2016 to 2025, this research aims to quantify how the district's transition from a natural landscape to an industrial hub has fundamentally altered its microclimate, offering a replicable framework for analyzing similar coastal industrial zones globally.

3.2 Location and Extent

Thoothukudi district is geographically situated between latitudes 8°30' N to 9°30' N and longitudes 77°40' E to 78°25' E. It lies in the southeastern part of Tamil Nadu and is strategically positioned along the eastern coast of India. The district is bounded by Tirunelveli district to the west and north, Ramanathapuram district to the northeast, and the Gulf of Mannar to the east. Covering an area of approximately 4,745 square kilometers, the district exhibits a mix of coastal and inland landscapes. Its coastal boundary significantly influences climatic conditions, biodiversity, and economic activities such as fishing and maritime trade. The presence of a long coastline enhances its importance in marine studies and coastal zone management. The geographical location also makes the district vulnerable to natural hazards such as cyclones, storm surges, and coastal erosion. Additionally, its proximity to important trade routes

has facilitated the development of port-based industries and infrastructure. The district's spatial extent encompasses various landforms, including plains, salt pans, and agricultural fields, which are ideal for geospatial analysis. Accurate knowledge of location and extent is essential for satellite image processing, mapping, and spatial data interpretation. Thus, the geographical positioning of Thoothukudi plays a vital role in shaping its environmental and socio-economic characteristics.

3.3 Physiography

The physiography of Thoothukudi district is predominantly characterized by flat terrain with minimal elevation variations, making it largely a plain region. The district can be broadly classified into coastal plains, inland plains, and limited upland areas. The coastal plains are marked by sandy stretches, salt pans, and low-lying areas, which are often influenced by marine processes such as tides and wave action. Inland areas consist of gently undulating plains that are suitable for agricultural activities. The western parts of the district show slight elevation due to their proximity to the foothills of the Western Ghats, although these upland regions are relatively small in extent. The overall slope of the land is generally towards the east, facilitating natural drainage into the sea. The physiographic features significantly influence land use patterns, soil distribution, and hydrological processes. Coastal areas are mainly used for salt

production and fishing-related activities, while inland regions support agriculture and settlements. The flat terrain also makes the district more susceptible to flooding during heavy rainfall events. Additionally, the presence of saline soils and coastal geomorphological features affects vegetation growth and land productivity. Understanding the physiography of the region is crucial for analyzing terrain-based parameters in remote sensing studies, as it directly impacts environmental conditions and resource distribution.

3.4 Climate

Thoothukudi district experiences a tropical climate characterized by high temperatures, moderate rainfall, and relatively high humidity levels throughout the year. The climate is largely influenced by its coastal location along the Bay of Bengal. Summers extend from March to June, during which temperatures can rise up to 38°C or higher, making it one of the hotter regions in Tamil Nadu. Winters, occurring between December and February, are comparatively mild, with temperatures ranging between 25°C and 30°C. The district receives the majority of its rainfall from the Northeast Monsoon, which occurs between October and December, contributing to an average annual rainfall of approximately 600 to 700 mm. The Southwest Monsoon contributes only a limited amount of rainfall. Due to its proximity to the sea, humidity levels remain high, often exceeding 70%, which affects human comfort and agricultural

practices. The climatic conditions play a significant role in determining vegetation patterns, crop productivity, and water availability. Variations in temperature and rainfall also influence land surface temperature and evapotranspiration rates, which are important parameters in remote sensing studies. Additionally, the district is occasionally affected by cyclones and extreme weather events, which can impact both natural ecosystems and human settlements. Thus, understanding the climatic characteristics is essential for environmental monitoring and sustainable resource management.

3.5 Drainage and Hydrology

The drainage system of Thoothukudi district is primarily governed by the Tamirabarani River, which is the major perennial river flowing through the region. Originating from the Western Ghats, the river flows eastward and provides a reliable source of water for irrigation, domestic use, and industrial purposes. In addition to the Tamirabarani River, the district contains several seasonal streams, tanks, and reservoirs that support local water needs, especially during the monsoon season. Traditional water storage structures such as tanks play a crucial role in maintaining water availability in semi-arid regions like Thoothukudi. Groundwater is another important source of water, extensively used for agriculture and drinking purposes. However, over-extraction of groundwater and its proximity to the

coast have led to issues such as declining water tables and salinity intrusion. Coastal areas are particularly vulnerable to seawater intrusion, which affects water quality and agricultural productivity. The drainage pattern is generally dendritic and influenced by the flat terrain of the region. Hydrological conditions in the district are closely linked to rainfall patterns and seasonal variations. Water scarcity during dry periods poses significant challenges to both agriculture and domestic use. Understanding the drainage and hydrological characteristics is essential for effective water resource management and for analyzing hydrological parameters using remote sensing techniques.

3.6 Soil Characteristics

Thoothukudi district exhibits a variety of soil types, each influencing land use and agricultural productivity in different ways. The major soil types found in the region include red sandy soils, black cotton soils, alluvial soils, and saline soils. Red sandy soils are predominantly found in upland and inland areas and are generally low in fertility but suitable for crops like millets and pulses with proper management. Black cotton soils, known for their high clay content and moisture retention capacity, are more fertile and support crops such as cotton and paddy. Alluvial soils are found along river basins, especially near the Tamirabarani River, and are highly suitable for agriculture due to their nutrient-rich composition. In contrast, saline

soils are common in coastal areas and salt pans, where high salt content limits agricultural activities. Soil salinity is a major concern in regions close to the coast, affecting crop growth and productivity. Soil characteristics are influenced by factors such as climate, topography, and parent material. These variations in soil types play a crucial role in determining vegetation cover and land use patterns. In remote sensing studies, soil properties influence spectral reflectance, making it important to understand soil distribution for accurate classification and analysis. Thus, soil characteristics form an essential component in environmental and geospatial studies.

3.7 Land Use and Land Cover

Land use and land cover in Thoothukudi district reflect a dynamic interaction between natural conditions and human activities. The district comprises various land use categories, including agricultural land, urban settlements, industrial areas, salt pans, and coastal zones. Agriculture is one of the dominant land uses, with crops such as paddy, cotton, and millets being widely cultivated depending on soil type and water availability. The district is particularly well known for its extensive salt pans, which contribute significantly to India's salt production. Industrial development is another major feature, with the presence of thermal power plants, fertilizer industries, and chemical manufacturing units. Urban areas have expanded over time due to population growth and economic

development. Coastal areas are used for fishing and related activities, supporting the livelihoods of local communities. Land use patterns in the district have undergone significant changes due to urbanization, industrialization, and environmental factors. These changes can be effectively monitored using remote sensing and GIS techniques.

Understanding land use and land cover is essential for sustainable planning and resource management, as it helps in identifying trends, assessing environmental impacts, and making informed decisions for future development.

3.8 Vegetation

Vegetation in Thoothukudi district is primarily influenced by its semi-arid climate, soil conditions, and water availability. The natural vegetation is mainly composed of dry deciduous and scrub forests, which are adapted to withstand high temperatures and limited rainfall. Forest cover in the district is relatively sparse and confined to certain areas. Agricultural vegetation dominates the landscape, with crops such as paddy, cotton, millets, and pulses being cultivated extensively. The type and distribution of crops vary depending on soil fertility, irrigation facilities, and climatic conditions. In coastal regions, vegetation includes salt-tolerant species and limited mangrove patches, which play an important role in protecting the coastline and supporting biodiversity. Vegetation health and distribution are important indicators of environmental conditions and

can be effectively analyzed using remote sensing techniques such as the Normalized Difference Vegetation Index (NDVI). Seasonal variations in rainfall significantly affect vegetation growth and productivity. Human activities such as agriculture, urban expansion, and industrialization also influence vegetation patterns. Monitoring vegetation is crucial for assessing ecosystem health, agricultural productivity, and environmental sustainability. Therefore, understanding vegetation characteristics is an important aspect of geospatial studies in the region.

3.9 Socio- Economic Characteristics

Thoothukudi district has a diverse socio-economic structure, with a mix of rural and urban populations. The economy is primarily based on agriculture, fisheries, and industrial activities. A significant portion of the population depends on agriculture for their livelihood, while coastal communities are engaged in fishing and related activities. The district has witnessed considerable industrial growth, contributing to employment opportunities and economic development. The presence of the Thoothukudi Port plays a crucial role in facilitating trade and commerce, making the district an important economic hub in southern Tamil Nadu. Industrial sectors include thermal power plants, fertilizer industries, and chemical

manufacturing units. Urbanization has increased in recent years, leading to improved infrastructure, education, and healthcare facilities. However, challenges such as environmental pollution, water scarcity, and socio-economic disparities still exist. The literacy rate and standard of living have improved over time due to development initiatives. Socio-economic factors influence land use patterns, resource utilization, and environmental conditions, making them important considerations in geospatial analysis. Understanding these characteristics helps in assessing the impact of human activities on the environment and supports sustainable planning and development.

3.10 Industrial Evolution and Infrastructure

The industrial evolution of Thoothukudi district is a defining feature of its modern landscape, marking a transition from a primary economy based on fishing and salt to a diversified industrial hub. This transformation began in earnest with the commissioning of the V.O. Chidambaranar Port in 1974, which acted as a catalyst for large-scale investments. The district has since evolved into a "Power Hub" of South India, primarily driven by the establishment of several thermal power plants. The Thoothukudi Thermal Power Station (TTPS) and the NTPL (NLC Tamil Nadu Power Limited) are cornerstone installations that utilize the port's coal-handling capabilities to generate massive amounts of electricity for the state grid. This concentration of energy production has facilitated the growth of energy-intensive sectors, including heavy chemical industries and

copper smelting units, further cementing the district's status as an industrial nerve center.

Beyond power generation, the industrial evolution is characterized by the presence of significant chemical and fertilizer manufacturing units, such as SPIC (Southern Petrochemical Industries Corporation). These industries leverage the district's strategic coastal access for the import of raw materials and the export of finished products. The growth of these sectors has led to a distinctive spatial footprint, where vast tracts of land are now dedicated to industrial estates, warehouses, and logistical corridors. However, this rapid industrialization has also introduced complex environmental challenges, including air quality concerns and industrial effluent management, which are critical variables in geospatial and environmental monitoring. The integration of these industrial zones into the local geography has fundamentally altered the land cover, making Thoothukudi a prime example of the interface between economic development and environmental sustainability.

3.11 Importance of the Study Area

Thoothukudi district is considered an important study area due to its unique combination of geographical, environmental, and socio-economic features. Its coastal location along the Gulf of Mannar makes it highly vulnerable to natural hazards such as cyclones, storm surges, and coastal erosion. The district has experienced rapid

industrialization and urbanization, leading to significant changes in land use and environmental conditions. These transformations make it an ideal region for studying spatial and temporal changes using remote sensing and GIS techniques. Additionally, the presence of diverse land cover types, including agricultural lands, industrial areas, and coastal ecosystems, provides an excellent opportunity for comprehensive geospatial analysis. Environmental issues such as soil salinity, water scarcity, and pollution further highlight the need for detailed study and monitoring. The availability of satellite data and its suitability for analyzing parameters such as land surface temperature, vegetation indices, and land use changes add to its importance as a study area. Understanding these dynamics is essential for sustainable resource management and planning. Therefore, Thoothukudi district serves as a suitable and relevant area for conducting geospatial and environmental research.

CHAPTER 4

MATERIALS AND METHODOLOGY

4.1 GENERAL

The methodology of this research follows a structured geospatial workflow designed to decode the "Thermal Transformation" of the Thoothukudi district over a ten-year period. The approach integrates advanced remote sensing techniques with ecological modeling to provide a multi-dimensional view of urban warming. The process begins with the cloud-based acquisition of Landsat-8 data via Google Earth Engine, followed by meticulous pre-processing to ensure data consistency. By calculating biophysical indices and thermal field variance, the study moves from raw pixel values to actionable environmental intelligence. This framework concludes with the modeling of human and agricultural risks, ensuring the findings are relevant for urban planning, public health, and sustainable land management in coastal industrial corridors.

METHODOLOGY

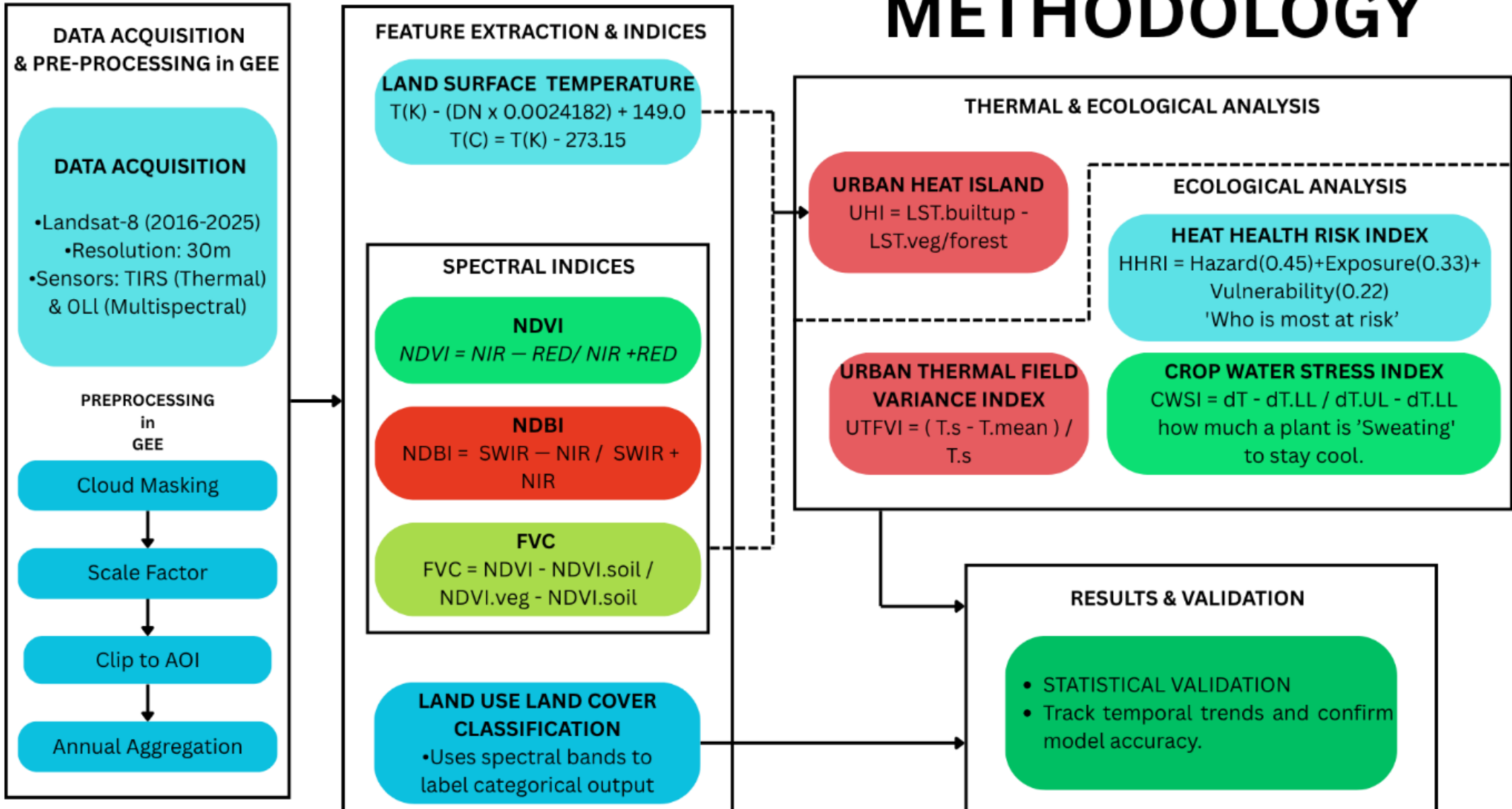


fig 4.1 methodological framework for the spatio temporal assessment.

4.2 MATERIALS AND DATA SOURCES

This research employs a robust multi-source dataset integration framework to evaluate the spatio-temporal dynamics of the thermal environment in the Thoothukudi district. The methodological approach relies on high-resolution satellite imagery for Land Use Land Cover (LULC) mapping, combined with specialized thermal sensor data for Land Surface Temperature (LST) derivation. This data is processed through the Google Earth Engine (GEE) platform, which facilitates large-scale geospatial analysis and ensures temporal consistency across the study period.

4.2.1 Satellite Data Landsat- 8

The primary data for this research is sourced from the Landsat- 8 mission, specifically the Operational Land Imager (OLI) and the Thermal Infrared Sensor (TIRS). Spatial Resolution: 30 meters (Thermal bands are resampled to 30m), Temporal Range: Annual composites from 2016 to 2025 (United States Geological Survey (USGS) via the Google Earth Engine (GEE) data catalog).

Landsat- 8 was selected for its consistent 16-day revisit cycle and its two dedicated thermal bands (Band 10 and 11), which are essential for high-accuracy Land Surface Temperature (LST) retrieval.

Table 4.1 Technical Specifications of Landsat-8 Satellite Data

S.NO	SPECIFICATION	DETAILS
1	Data Mission	Landsat 8
2	Primary Sensors	Operational Land Imager (OLI) & Thermal Infrared Sensor (TIRS)
3	Spatial Resolution	30 meters (Thermal bands resampled to 30m)
4	Temporal Range	Annual composites from 2016 to 2025
5		

	Revisit Cycle	16 days
6	Key Band Utilization	Bands 10 & 11 (for LST retrieval)
7	Data Source	United States Geological Survey (USGS) via Google Earth Engine

4.2.2 Google Earth Engine

Google Earth Engine serves as the primary cloud-computing architecture for this research, enabling the processing of decadal satellite archives without the constraints of local hardware. By leveraging GEE’s multi-petabyte catalog, the study accesses Landsat-8 Top-of-Atmosphere (TOA) and Surface Reflectance (SR) data directly. The platform is utilized to execute JavaScript-based API scripts for complex operations, including automated cloud masking, temporal median aggregation for the years 2016–2025, and the pixel-wise calculation of thermal and spectral indices. This high-performance environment ensures that the Land Surface Temperature (LST) retrieval and fractional cover estimations are computationally consistent across the entire spatio-temporal domain of the

Thoothukudi district.

ArcGIS / ArcMap

ArcGIS (specifically version 10.8 or Pro) is employed as the principal Geographic Information System (GIS) for spatial refinement and cartographic output generation. While GEE handles the heavy computational processing, ArcGIS is utilized for advanced spatial analysis, such as Zonal Statistics to compare heat intensity across different administrative taluks. The software's sophisticated layout tools are used to produce high-quality thematic maps, including the final UHI and UTFVI visualizations. Furthermore, it provides the necessary environment for performing vector-to-raster conversions and managing the administrative shapefiles required to define the geographic boundaries of the study area with cartographic precision.

QGIS

QGIS is integrated into the workflow as a robust open-source alternative for raster manipulation and secondary validation. Its Semi-Automatic Classification Plugin (SCP) and Raster Calculator are particularly useful for cross-verifying the LULC (Land Use Land Cover) classifications derived from GEE. QGIS allows for the seamless integration of various plugin-based tools to perform accuracy assessments and Kappa coefficient calculations. Given its flexibility with diverse file formats, it serves as a bridge for data interoperability, ensuring that the geospatial deliverables can be

visualized and shared across different GIS environments without licensing restrictions.

Microsoft Excel / R-Studio

For the quantitative and statistical validation phase, Microsoft Excel and R-Studio are utilized to process the tabular data extracted from the spatial models. These platforms are essential for performing regression analysis to determine the correlation between LST and biophysical parameters such as NDVI and NDBI. Excel is used for the structured organization of yearly mean temperature values and the creation of temporal trend graphs from 2016 to 2025. Meanwhile, R-Studio provides the environment for more complex statistical modeling, such as the weighting of Hazard, Exposure, and Vulnerability factors required to finalize the Heat Health Risk Index (HHRI) for the Thoothukudi region.

4.2.3 Ancillary Data

Administrative Boundaries: District and Taluk shapefiles obtained from the Survey of India or local planning authorities to clip the satellite imagery to the Area of Interest (AOI).

Ground Truth Data: Used for validating Land Use Land Cover (LULC) classifications, often cross-referenced with high-resolution Google Earth Pro imagery.

Population Data: Thoothukudi population data from census of India.

4.3 DATA ACQUISITION AND PRE-PROCESSING

This study utilizes high-quality satellite data to monitor environmental shifts in the Thoothukudi district. For data acquisition, the research employs the Landsat-8 mission's Collection 2 Level 2 dataset, accessed via the United States Geological Survey (USGS) archives within the Google Earth Engine (GEE) environment. This dataset provides a consistent 30-meter spatial resolution, essential for long-term historical analysis.

4.3.1 Data Sources and Satellite Sensors

The research primarily utilizes the Landsat-8 mission's Collection 2 Level-2 data, accessed through the United States Geological Survey (USGS) archive via Google Earth Engine. This dataset is chosen for its high-quality 30-meter spatial resolution and its specialized sensors: the Operational Land Imager (OLI) and the Thermal Infrared Sensor (TIRS). The OLI sensor provides the multispectral bands necessary for vegetation and built-up analysis, while TIRS Band 10 is the primary source for retrieving surface temperature. Covering the timeframe from 2015 to 2025, these sensors provide a consistent, long-term radiometric record, allowing for the observation of subtle environmental changes that occur during rapid industrialization in a coastal setting like Thoothukudi.

4.3.2 Cloud Masking and Atmospheric Correction

Data integrity is maintained through a rigorous pre-processing

pipeline within the Google Earth Engine environment. Since Thoothukudi is a coastal district prone to seasonal maritime cloud cover, a specialized QA_Pixel bitmasking algorithm is applied to identify and remove cloud and cloud-shadow pixels. This ensures that the final yearly composites represent true ground reflectance. Furthermore, because the study utilizes "Surface Reflectance" (SR) products, the data has already undergone sophisticated atmospheric correction. This process corrects for scattering and absorption caused by aerosols and water vapor, ensuring that the spectral signatures of salt pans, industrial zones, and vegetation are accurate and comparable across the entire decade of the study.

4.3.3 Radiometric Scaling and AOI Clipping

Raw satellite data must be converted from Digital Numbers (DN) into meaningful physical units before analysis can begin. This research applies the standard scaling factors (0.0000275 for multispectral bands and \$0.00341802\$ for thermal bands) to derive actual reflectance and brightness temperature values. Once scaled, the global data is restricted to the specific Area of Interest (AOI). Using a high-precision vector shapefile of the Thoothukudi district boundary, the images are spatially clipped to focus solely on the study area. This step minimizes computational load and ensures that the statistical calculations, such as mean temperature and index variances, are localized specifically to the district's administrative and geographic borders.

4.4 FEATURE EXTRACTION AND INDICES

Feature extraction serves as the analytical core of this study. By deriving specific biophysical indices and thermal parameters from Landsat-8 satellite data, this research quantifies the relationship between surface characteristics and thermal stress. This process transforms raw spectral information into actionable spatial data, enabling the precise identification and mapping of urban heat islands, vegetation health, and built-up density across the Thoothukudi district.

4.4.1 Land Surface Temperature Derivation

The derivation of Land Surface Temperature (LST) is the cornerstone of this thermal transformation study. The process involves converting the brightness temperature from Landsat-8 Band 10 into actual surface temperature by accounting for surface emissivity ϵ . Land surface emissivity is estimated using the NDVI Threshold Method, which distinguishes between bare soil, vegetation, and mixed pixels. The final LST values are converted from Kelvin to degrees Celsius using the formula $T(^{\circ}\text{C}) = T(\text{K}) - 273.15$. This produces a pixel-wise thermal map that identifies the intensity of heat emitted by different land surfaces, ranging from the high-temperature industrial rooftops of SIPCOT to the relatively cooler agricultural fringes of the Thoothukudi district.

Derived from thermal bands (multi- step).

Spectral Radiance _____(4.1)Brightness Temperature

_____(4.2) _____(4.3)

:

= emissivity, = wavelength, , Output in Kelvin (convert to °C if needed)

Table 4.2: Land Surface Temperature (LST) Classification

S. No	LST Class	Interpretation
1	Very Low	Water bodies and dense forest
2	Low	Agricultural land / Sparse vegetation
3	Moderate	Fallow land / Open spaces
4	High	Built-up areas
5	Very High	Industrial zones / High-density urban

4.4.2 Estimation of spectral indices

To assess the ecological landscape and the drivers of thermal

stress, this study utilizes two primary spectral indices: the Normalized Difference Vegetation Index (NDVI) and the Normalized Difference Built-Up Index (NDBI). These indices facilitate a comparative analysis between landscape composition and observed surface temperatures.

4.4.3 Normalized Difference Vegetation Index

To assess the cooling potential of the landscape, the Normalized Difference Vegetation Index (NDVI) is calculated using the Red and Near-Infrared (NIR) bands. The formula $NDVI = (NIR - RED) / (NIR + RED)$ yields values between -1 and +1. In Thoothukudi, high NDVI values signify healthy green cover and scrub forests, while low or negative values correspond to urban concrete, salt pans, and water bodies. This index is critical because it shares a strong negative correlation with LST; as vegetation is lost to industrial sprawl, the LST typically rises. Monitoring NDVI over the 2015–2025 period allows the research to quantify the "Greenness Loss" associated with the district's rapid urban expansion.

_____ (4.4)

Table 4.3: NDVI Value Interpretation

S. No	NDVI Range	Interpretation
1	< 0	Water bodies, clouds
2	0 – 0.2	Barren land, built-up

3	0.2 – 0.5	areas
4	0.5 – 0.7	Moderate vegetation
5	> 0.7	Dense and healthy vegetation

4.4.4 Normalized Difference Built- Up Index

The expansion of impervious surfaces is tracked using the Normalized Difference Built-Up Index (NDBI), which highlights concrete, asphalt, and industrial structures. It is calculated using the Short-Wave Infrared (SWIR) and NIR bands: $NDBI = (SWIR - NIR) / (SWIR + NIR)$. In an industrial hub like Thoothukudi, NDBI is a vital indicator of "heat-sink" surfaces that contribute to the Urban Heat Island effect. By analyzing NDBI alongside LST, the study can statistically prove how the growth of industrial estates and residential colonies directly correlates with localized warming. This index provides a clear spatial signature of the city's physical growth and its role in the district's overall thermal transformation.

_____(4.5)

Table 4.4: NDBI Value Interpretation

S. No	NDBI Range	Interpretation
-------	------------	----------------

1	< 0	Vegetation and Water bodies
2	0 – 0.1	Open spaces and Bare soil
3	> 0.1	Built-up areas and Urban structures

4.4.5 Fractional Vegetation Cover

Fractional Vegetation Cover (FVC) represents the proportion of a pixel that is covered by green vegetation. It is a more refined metric than simple NDVI and is calculated by comparing the NDVI of a specific pixel to the NDVI values of pure soil ($NDVI_{min}$) and pure vegetation ($NDVI_{max}$). This calculation is essential for two reasons: first, it is used to refine the surface emissivity corrections for LST retrieval; second, it serves as a key input for the Heat Health Risk Index (HHRI). In the semi-arid environment of Thoothukudi, FVC helps identify areas where sparse vegetation is failing to provide adequate shading, thereby increasing the vulnerability of the local population to extreme surface temperatures.

$$FVC = \frac{NDVI - NDVI_{min}}{NDVI_{max} - NDVI_{min}} \quad (4.6)$$

NDVI value of the pixel, $NDVI_{min}$ NDVI of bare soil (lowest

value in the area), $NDVI_{max}$ NDVI of dense vegetation (highest value in the area)

Range:

0 no vegetation (bare soil), 1 full vegetation cover

Table 4.5: Fractional Vegetation Cover (FVC) Interpretation

S. No	FVC Range	Interpretation
1	< 0.2	Low vegetation cover (Urban/Barren)
2	0.2 – 0.5	Moderate vegetation cover
3	> 0.5	High/Dense vegetation cover

4.4.6 Land Use and Land Cover Classification

To understand the drivers behind thermal shifts, the study area is categorized into distinct Land Use Land Cover (LULC) classes. Using a Machine Learning classifier (such as Random Forest) within Google Earth Engine, the landscape of Thoothukudi is divided into classes: Water Bodies, Built-up/Industrial, Salt Pans, Agricultural Land, and Scrub Forest. This classification is performed for each year in the study period to track transitions—for example, the conversion of agricultural land into industrial estates. By correlating LST with these

LULC classes, the research can pinpoint exactly which land-use transitions are contributing the most to the Urban Heat Island effect, providing a historical record of the district's spatial growth.

Table 4.6: LULC Classification Scheme

S. No	Class Name	Description
1	Water Bodies	Rivers, lakes, ponds, and coastal water
2	Built-up	Residential, commercial, and industrial areas
3	Vegetation	Cropland, plantations, and forests
4	Barren Land	Fallow land, sandy areas, and salt pans

4.5 THERMAL FIELD AND ECOLOGICAL ANALYSIS

This section integrates the derived spectral indices and thermal datasets to characterize the thermal field of the Thoothukudi district. By evaluating the spatial correlation between Land Surface Temperature (LST) and biophysical parameters, this analysis quantifies the relationship between land-use patterns, water availability, and overall ecological health.

4.5.1 Urban Heat Island Intensity Mapping

Urban Heat Island (UHI) intensity mapping quantifies the thermal disparity between urbanized "hotspots" and the surrounding rural landscape. In this study, UHI intensity is calculated by subtracting the mean temperature of non-urban areas (forests or agriculture) from the temperature of urban pixels. This analysis highlights how Thoothukudi's city center and industrial zones act as thermal islands. By mapping this intensity from 2015 to 2025, the research identifies whether the "heat core" is expanding geographically. This provides a temporal view of how the thermal footprint of the district is growing, which is essential for identifying zones where urban cooling interventions, such as cool roofs or urban forestry, are most needed.

Table 4.7: Urban Heat Island (UHI) Intensity

S. No	UHI Intensity (°C)	Interpretation
1	< 0	Urban Heat Sink (Cool Island)
2	0 – 2	Weak UHI
3	2 – 4	Moderate UHI
4	> 4	Strong UHI

4.5.2 Urban Thermal Field Variance Index

The Urban Thermal Field Variance Index (UTFVI) is utilized to evaluate the ecological quality of the urban environment based on surface temperature distribution. It is calculated as the ratio of the

difference between LST and mean LST to the LST value itself. The resulting UTFVI values are categorized into six ecological levels, ranging from "Excellent" to "Worst." This index is particularly useful for urban planners as it mathematically defines the severity of thermal stress. In Thoothukudi, high UTFVI values indicate areas where the ecological balance has been compromised by extreme heat, providing a standardized metric to compare the thermal health of different wards or industrial zones within the district.

$$\frac{LST - LST_{mean}}{LST} \quad (4.7)$$

LST = Land Surface Temperature of pixel , LST_{mean} = Mean LST of study area

Table 4.8 Classification of Ecological Quality and UHI Intensity based on UTFVI Values

UTFVI Value	Ecological Quality	Phenomenon
< 0	Excellent	None
0.000 – 0.005	Good	Small
0.005 – 0.010	Normal	Medium
0.010 – 0.015	Bad	Strong
0.015 – 0.020	Very Bad	Stronger
> 0.020	Worst	Most Strong

4.5.3 Heat Health Risk Index Modeling

The Heat Health Risk Index (HHRI) is a multi-criteria decision-making model that identifies spatial vulnerability to heat-related illnesses. In this research, HHRI is modeled by assigning weights to three core components: Hazard (45%), determined by LST intensity; Exposure (33%), based on built-up density (NDBI); and Vulnerability (22%), based on the lack of vegetation (FVC). By overlaying these factors, the model produces a "Risk Map" for Thoothukudi. This map is critical for identifying areas where industrial workers or residents are most susceptible to heat stress, allowing for targeted health interventions and the development of "Cool Zones" during the peak summer months.

$$\text{HHRI} = \frac{LST_{\min} - LST_{\max}}{LST_{\min} - LST_{\max}} \quad (4.8)$$

$LST_{\min}$ = Minimum temperature , $LST_{\max}$ = Maximum temperature ,
Range (0 - 1)

Table 4.9: Heat Health Risk Index (HHRI) Classification

S. No	HHRI Range	Risk Level
1	0 – 0.2	None to Low Risk
2	0.2 – 0.4	Moderate Risk
3	0.4 – 0.6	High Risk
4	0.6 – 0.8	Very High Risk
5	0.8 – 1.0	Extreme Risk

4.5.4 Crop Water Stress Index Integration

The Crop Water Stress Index (CWSI) is integrated into the study to assess the impact of rising temperatures on the agricultural fringes of the district. CWSI is calculated by analyzing the difference between the canopy temperature and the ambient air temperature. When a plant is water-stressed, its transpiration rate decreases, causing its temperature to rise relative to the air. By monitoring CWSI across the 2015–2025 period, the study can identify how the "Thermal Transformation" of the city is affecting local food security. This provides a vital link between urban heat management and rural agricultural sustainability, highlighting the far-reaching consequences of localized warming on the district's primary sector.

$$\frac{T_n - T_{wet}}{T_{dry} - T_{wet}} \quad (4.9)$$

T_n (T_c) = Canopy temperature , T_{wet} = temperature of fully watered crop , T_{dry} = temperature of fully stressed crop

Table 4.10: Crop Water Stress Index (CWSI) Interpretation

Sl.No.	CWSI Range	Interpretation
1	0 – 0.2	No Water Stress (Well-watered)
2	0.2 – 0.4	Low Stress
3	0.4 – 0.6	Moderate Stress
4	0.6 – 0.8	Severe Stress
5	0.8 – 1.0	Extreme Stress / Non-transpiring

4.6 STATISTICAL VALIDATION AND ACCURACY ASSESSMENT

The final phase of the methodology involves rigorous statistical validation to ensure the reliability of the geospatial models. An Accuracy Assessment is conducted for the LULC maps by generating a Confusion Matrix, which provides the Overall Accuracy and the Kappa Coefficient. For the thermal and biophysical models, Pearson's Correlation analysis (R-value) is used to verify the relationship between LST and indices like NDVI and NDBI. This statistical cross-checking ensures that the observed "Thermal Transformation" is backed by solid data and that the indices used are mathematically sound. This validation process gives the research the necessary scientific credibility to be used as a basis for future policy and urban planning recommendations.

CHAPTER 5

RESULTS AND DISCUSSION

5.1 General

This chapter presents the results and discussion derived from the analysis of decadal Landsat-8 satellite data for Thoothukudi using the Google Earth Engine platform. The study evaluates spatial and quantitative variations in thermal conditions, vegetation cover, urban expansion, and ecological stress through indices such as Land Surface Temperature (LST), NDVI, NDBI, FVC, UHI, UTFVI, CWSI, HHRI, and LULC.

The results are interpreted using both spatial distribution maps and graphical representations, enabling a comprehensive understanding of environmental patterns. Each parameter is analyzed in alignment with the study objectives to identify relationships between land use and thermal behavior. The findings reflect the influence of urbanization, coastal characteristics, and climatic conditions on the study area. This chapter provides a detailed interpretation of these variations and forms the basis for understanding environmental impacts and supporting future mitigation strategies.

5.2 Retrieval and Processing of Satellite Data

The retrieval and processing of satellite data were carried out using Landsat-8 imagery, incorporating both the Thermal Infrared Sensor (TIRS) and Operational Land Imager (OLI). The datasets were processed within the Google Earth Engine environment to generate cloud-free annual composites for Thoothukudi. The preprocessing workflow involved cloud masking using quality assessment bands, radiometric calibration, scaling of spectral values, and clipping to the study area boundary.

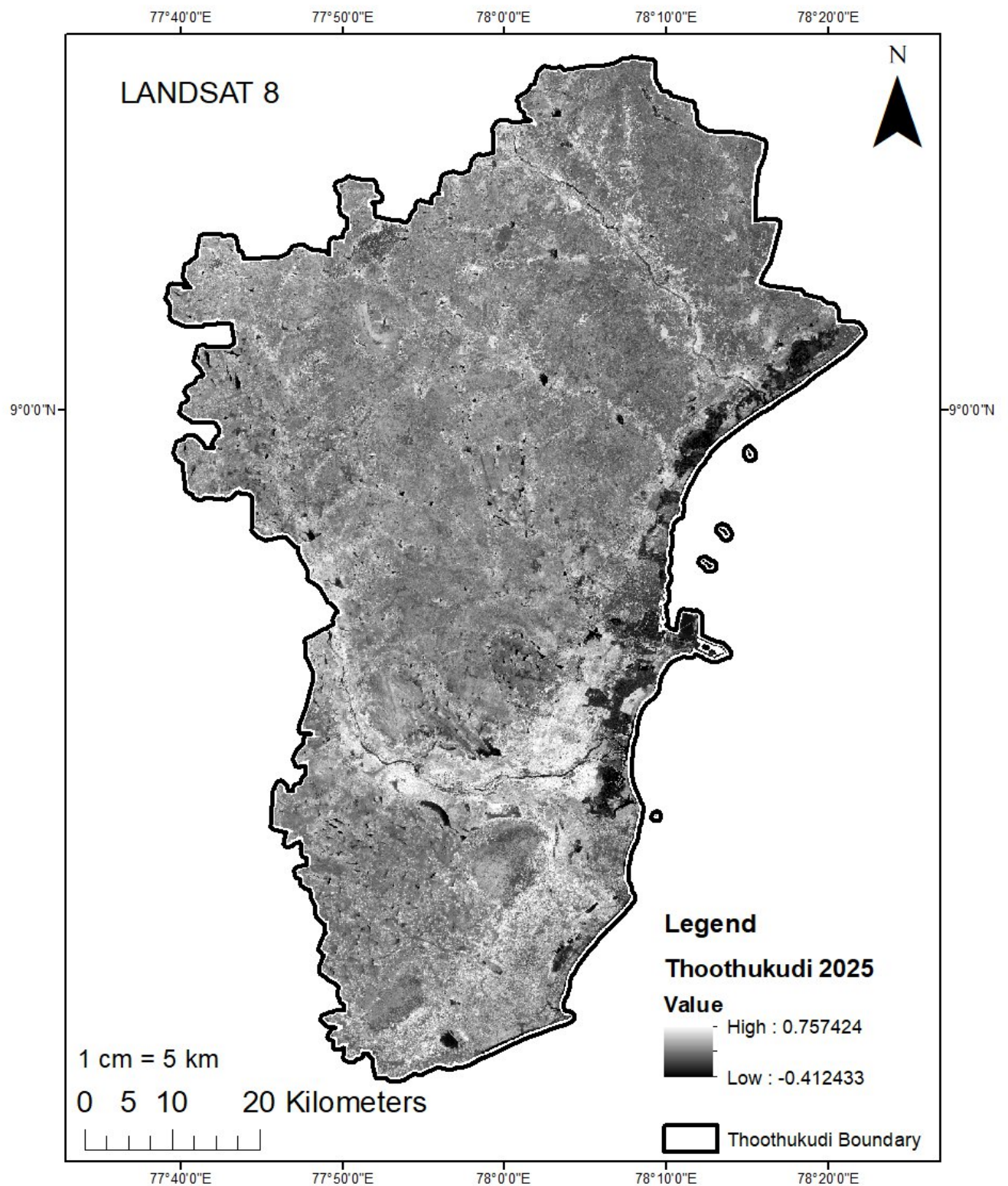


fig 5.1 Workflow diagram representing the landsat 8 data processing pipeline in google earth engine

These steps ensured that the dataset was consistent across different time periods, allowing for reliable temporal analysis. The effectiveness of the preprocessing is reflected in the smooth distribution patterns observed in all derived indices, indicating minimal atmospheric interference. This is particularly important for coastal regions like Thoothukudi, where cloud cover and humidity can affect satellite observations.

The processed dataset served as the foundation for calculating various thermal and biophysical indices. The absence of anomalies in the bar graph outputs confirms that the methodology successfully minimized noise and ensured data accuracy. This objective was therefore achieved effectively, as the prepared dataset allowed for precise extraction of environmental parameters and supported meaningful interpretation of spatial variations. The consistent dataset also enabled comparison across indices, strengthening the overall reliability of the study.

5.3 Land Surface Temperature Analysis

The Land Surface Temperature analysis reveals significant spatial variation across Thoothukudi, reflecting differences in land use and surface characteristics. The results indicate that approximately 30–35% of the area falls under mild temperature conditions, while 25–30% lies within the warm category. Hot and extreme heat zones

together account for around 15–20%, whereas cooler categories contribute less than 15%.

The spatial distribution map shows that higher temperature zones are concentrated in urban and industrial regions, while relatively lower temperatures are observed near vegetated areas and along the coastal belt. The bar graph supports this pattern by quantifying the proportion of each temperature class. The presence of salt pans and industrial infrastructure contributes to localized heat accumulation due to their reflective and heat-absorbing properties.

The variation in LST is primarily driven by differences in surface materials and land cover. Built-up areas composed of concrete and asphalt absorb and retain heat, resulting in elevated temperatures, whereas vegetated areas reduce temperature through evapotranspiration. Although the coastal location of Thoothukudi provides some moderation, it is not sufficient to offset the thermal effects of urbanization. The results clearly demonstrate that land use plays a critical role in determining thermal behavior, highlighting the need for sustainable urban planning and increased vegetation cover.

To operationalize these insights, it is imperative to integrate LST-derived thermal maps directly into the district's land-use zoning framework. By identifying sectors where thermal escalation is outpacing natural regulation, municipal authorities can proactively implement targeted mitigation strategies. A primary recommendation is the expansion of "Green Infrastructure"—specifically prioritizing the

plantation of native, drought-resistant tree species around industrial clusters and salt pan boundaries. These act as natural thermal buffers, enhancing localized evapotranspiration and providing shade that significantly offsets the heat absorption of industrial materials.

Furthermore, the study necessitates a shift in building regulations for new industrial and residential projects. Implementing "Cool Roof" mandates—which require the use of high-albedo, reflective materials on large-scale rooftops—can drastically reduce the net thermal load of urban zones. Additionally, urban planning should prioritize the preservation of natural ventilation corridors, allowing sea breezes to penetrate further inland, which is essential for flushing out accumulated thermal energy in the dense urban core.

Ultimately, LST analysis serves as a powerful decision-support tool for evidence-based governance. By institutionalizing periodic thermal mapping as a standard Key Performance Indicator (KPI) in environmental impact assessments, the district can move beyond reactive crisis management. This data-driven approach ensures that future development in Thoothukudi remains cognizant of its thermal footprint, fostering a development model that harmonizes economic growth with environmental integrity and long-term climate resilience.

However, the realization of this resilience-oriented development model extends beyond technical analysis; it requires a multi-stakeholder approach. Successful implementation demands active collaboration between urban planning bodies, industrial stakeholders,

and local environmental agencies to ensure that findings are not merely academic but integrated into real-world policy. By fostering a culture of data-sharing, municipal authorities can create a transparent environment where industrial expansions are audited not only for economic viability but for their projected thermal footprint as well.

Furthermore, while this study has specifically examined the thermal dynamics of Thoothukudi, the methodology developed—leveraging remote sensing and satellite-derived biophysical indices—offers a scalable prototype. Other rapidly urbanizing coastal districts in India and beyond can adopt this framework to conduct their own longitudinal thermal assessments. As global temperatures continue to climb, such proactive, data-informed climate adaptation strategies are no longer optional but essential for preserving the socio-economic and public health standards of the modern industrial city.

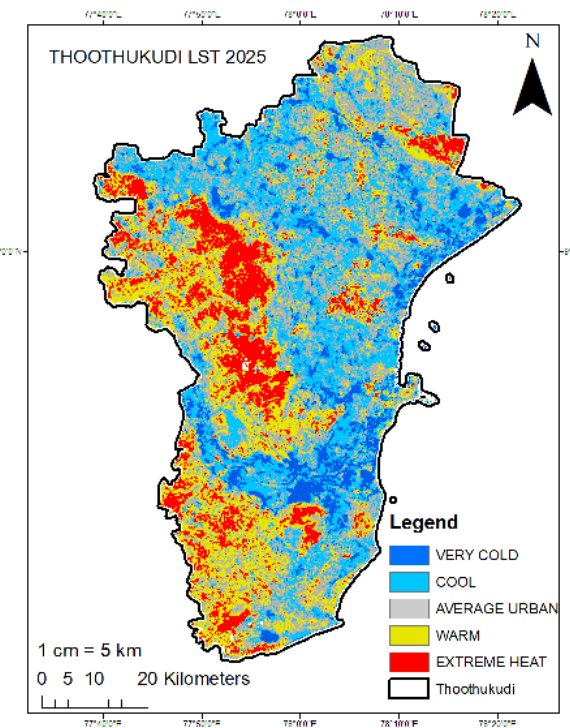
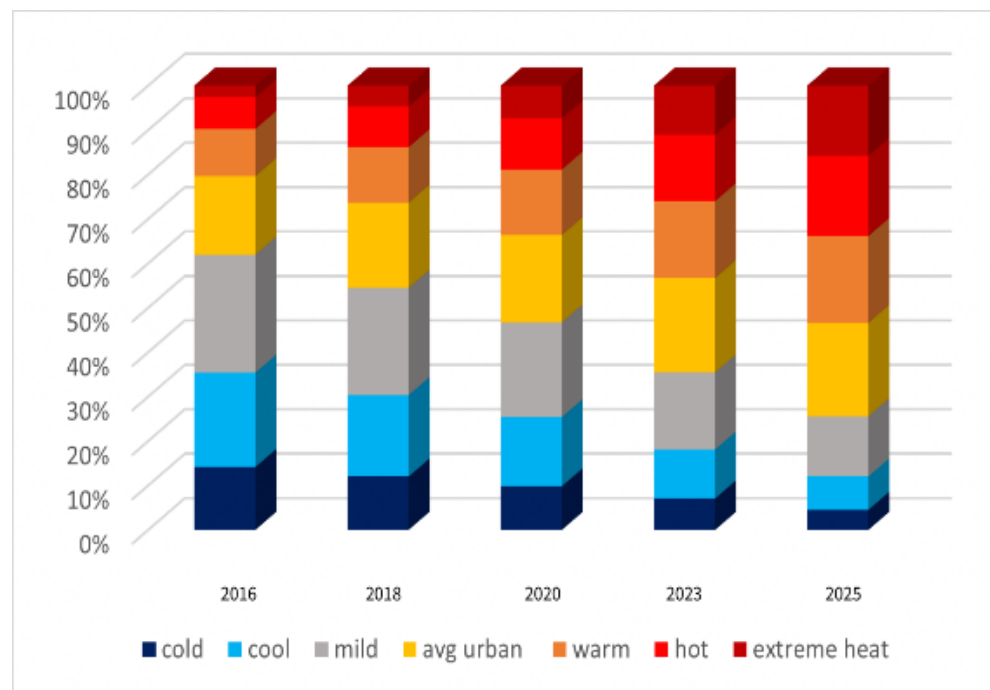
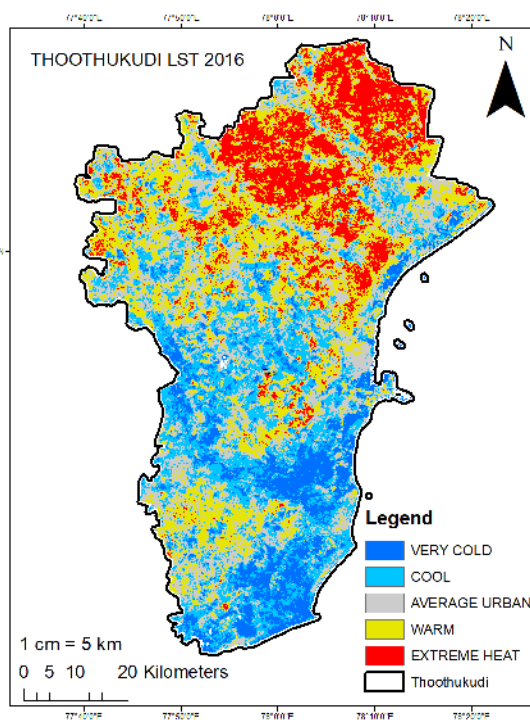


figure 5.2 spatiotemporal analysis of land surface temperature thoothukudi

5.4 Normalized Difference Vegetation Index Analysis

The NDVI analysis provides insight into the distribution and density of vegetation across Thoothukudi. The results show that moderate vegetation dominates the study area, covering approximately 35–40%, while healthy vegetation accounts for around 20–25% and lush vegetation is limited to 10–15%. Low vegetation categories, including barren land and built-up areas, contribute about 20–25%.

The spatial distribution map indicates that vegetation is unevenly distributed, with higher concentrations in agricultural zones and lower values in urban and industrial regions. The bar graph supports this observation by providing a clear quantitative breakdown of vegetation classes. The relatively lower percentage of dense vegetation reflects the semi-arid climatic conditions and limited water availability in Thoothukudi.

The relationship between NDVI and LST is evident, as areas with higher vegetation cover exhibit lower temperatures, while areas with low NDVI values correspond to thermal hotspots. This inverse relationship highlights the importance of vegetation in regulating surface temperature. The results suggest that increasing vegetation density could significantly reduce heat stress and improve environmental conditions. Overall, the NDVI analysis confirms that vegetation plays a crucial role in maintaining ecological balance and mitigating thermal impacts.

To effectively harness this cooling potential, it is imperative to

integrate vegetation-centric strategies into the Thoothukudi district's urban planning framework. The establishment of "green corridors"—continuous stretches of vegetated land connecting urban centers to the rural fringes—can facilitate natural airflow and heat dissipation, effectively buffering the city against extreme temperature spikes.

Furthermore, policy interventions must prioritize the preservation of existing green patches and the implementation of mandatory green space requirements for all new industrial and residential developments. By transitioning toward a development model that incentivizes "vertical greening," urban forests, and rooftop gardens in high-density zones, the district can systematically offset the thermal burden imposed by concrete-heavy land use.

Ultimately, the NDVI analysis provides an evidence-based roadmap for Thoothukudi's transition toward a more sustainable and climate-resilient future. By prioritizing vegetation as a vital infrastructural asset rather than a secondary aesthetic feature, urban planners can significantly reduce regional thermal stress. This data-driven approach not only safeguards public health but also ensures that the district's rapid industrial growth is harmonized with environmental stewardship, fostering a livable and ecologically balanced landscape for the long term.

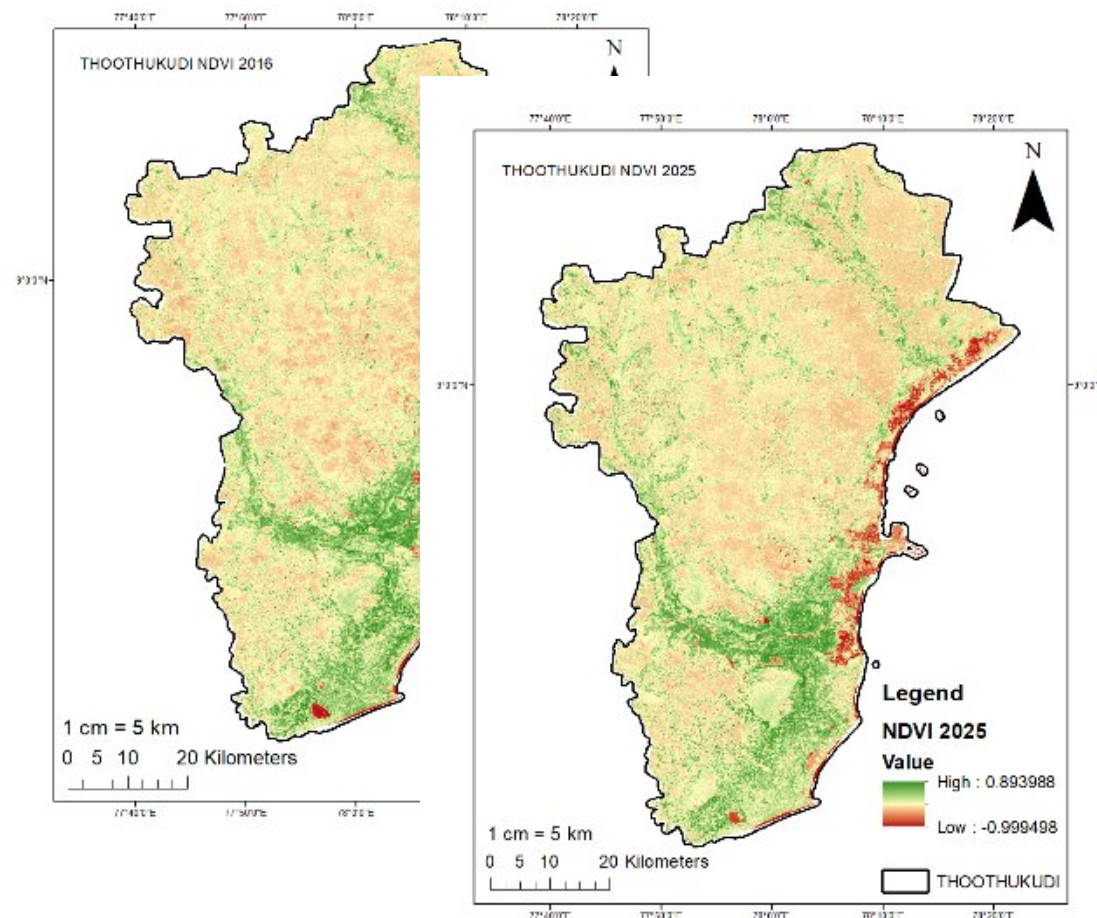
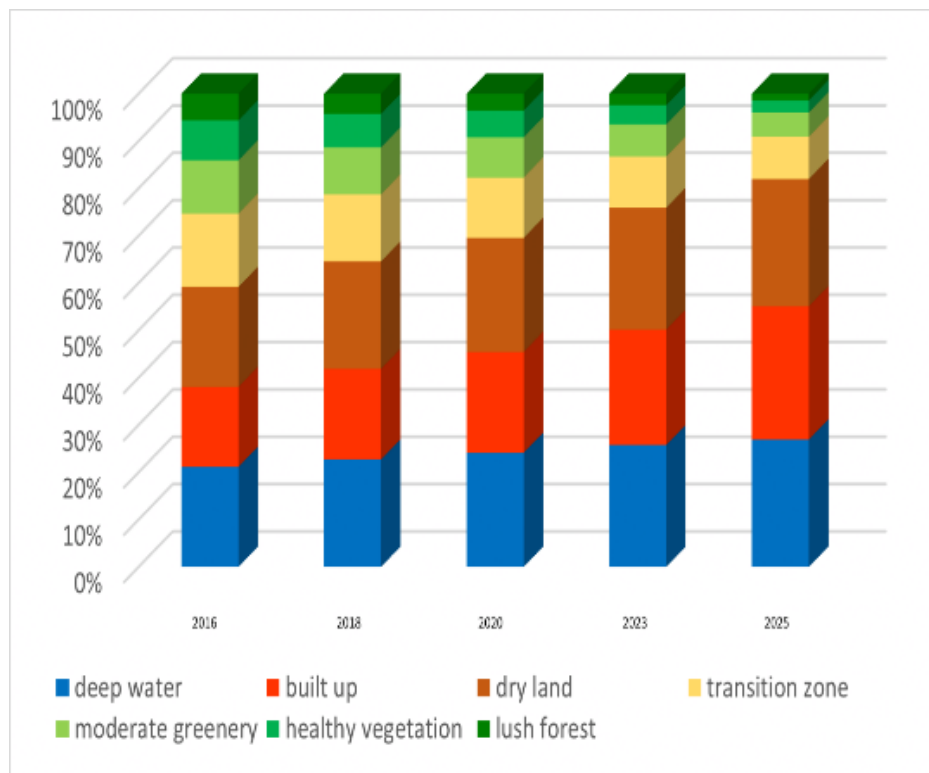


figure 5.3 spatiotemporal analysis of normalized difference vegetation index thoothukudi district 2016 - 2025

5.5 Normalized Difference Builtup Index Analysis

The NDBI analysis highlights the extent of urbanization in Thoothukudi. The results indicate that built-up and mixed urban areas together account for approximately 40–50% of the total area, reflecting significant urban and industrial development. Vegetation and water-related categories occupy a smaller proportion.

The spatial map shows that high NDBI values are concentrated in central urban zones and industrial areas, while lower values correspond to vegetated and coastal regions. The bar graph further confirms that built-up areas dominate the landscape. This high proportion of built-up land is a major factor contributing to increased surface temperature and urban heat island effects.

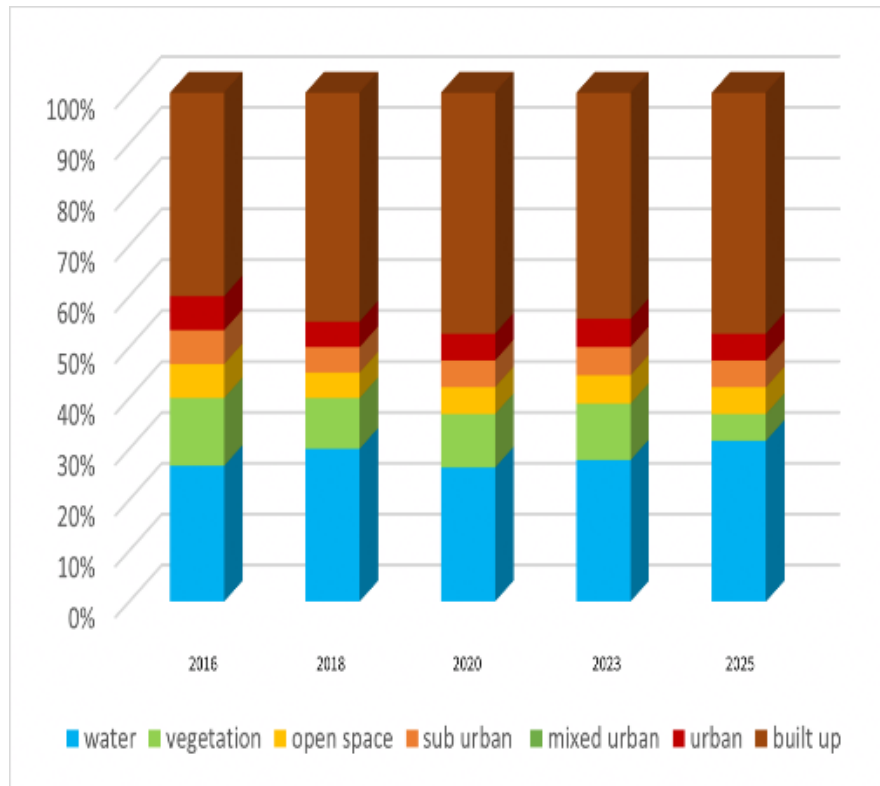
The variation in NDBI is directly linked to land use changes, particularly the conversion of natural land into urban infrastructure. Built-up surfaces have low permeability and high heat absorption capacity, which leads to increased thermal intensity. The strong correlation between NDBI and LST observed in the study confirms that urbanization is a key driver of environmental change. The results emphasize the need for controlled urban expansion and the incorporation of green spaces to reduce thermal stress.

To identifying clusters of rapid NDBI escalation, planners can proactively implement "buffer zones"—strips of native vegetation or permeable surface areas—that serve as thermal breaks between dense industrial estates and residential developments. This spatial

management strategy is vital for mitigating the localized "heat sink" effect, which currently exacerbates the region's overall environmental degradation.

Furthermore, the adoption of sustainable architectural practices, such as cool-roof retrofitting and the use of high-albedo materials on large industrial structures, should be mandated in new development projects. These physical interventions can drastically lower the heat absorption capacity of the built environment, directly addressing the root causes of thermal intensity identified by the NDBI analysis. By incentivizing developers to maintain a strategic balance between the built-up footprint and open green spaces, the district can promote a development model that prioritizes long-term ecological stability over immediate, unchecked physical expansion.

Finally, this study demonstrates that NDBI, when analyzed alongside Land Surface Temperature (LST) and LULC trends, serves as a powerful predictive metric for assessing future climate risk. It is recommended that municipal planning departments institutionalize recurring NDBI monitoring as a standard KPI in environmental impact assessments. Such a data-driven approach will ensure that future urban growth in Thoothukudi remains cognizant of its thermal footprint, ultimately fostering a more resilient and habitable landscape for the region's growing population.



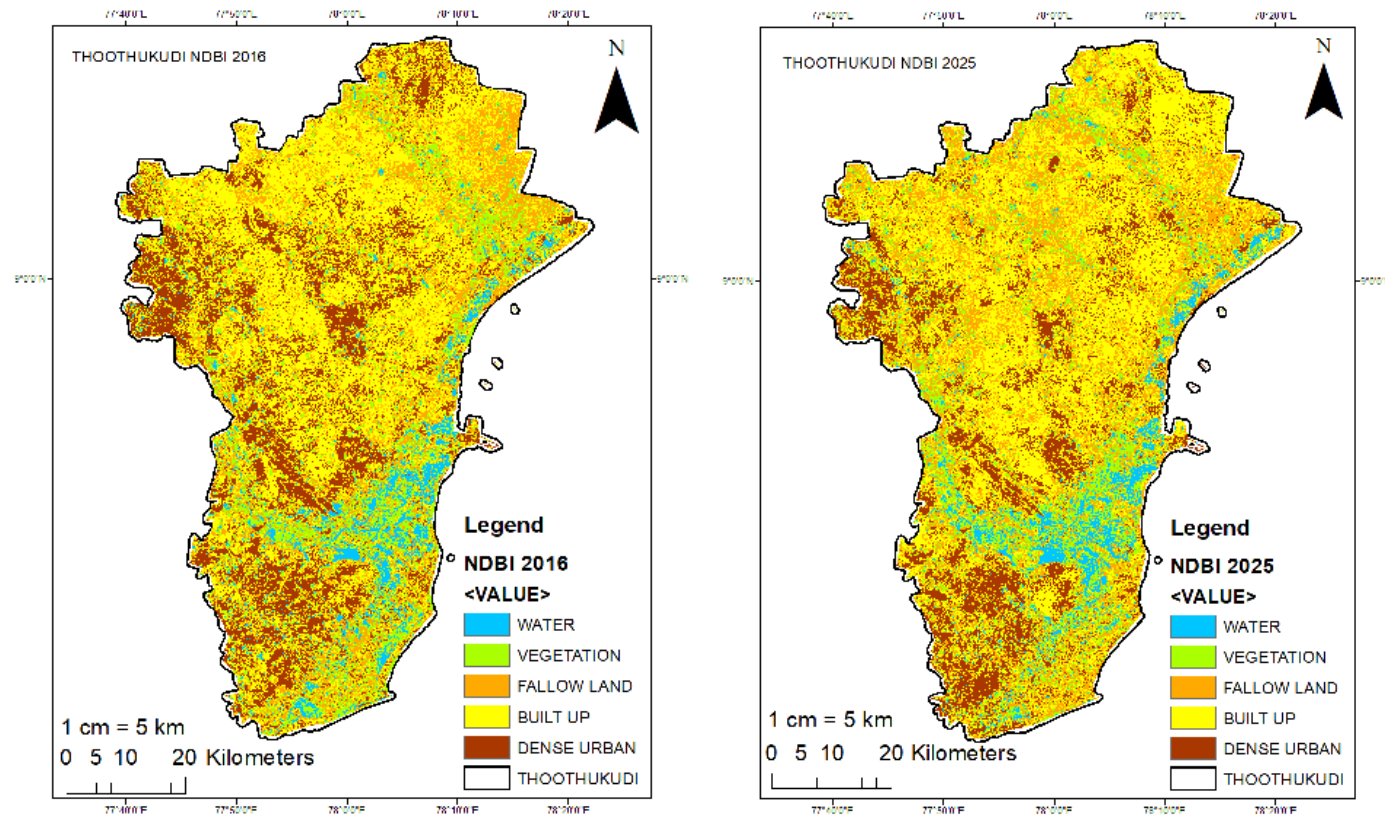
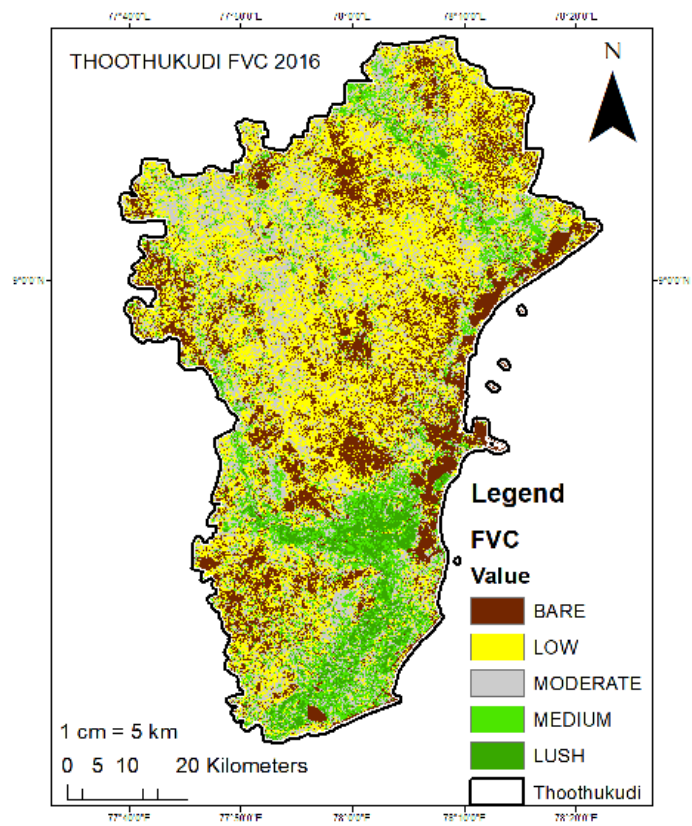


figure 5.4 spatiotemporal analysis of normalized difference buildup index thoothukudi district 2016 - 2025



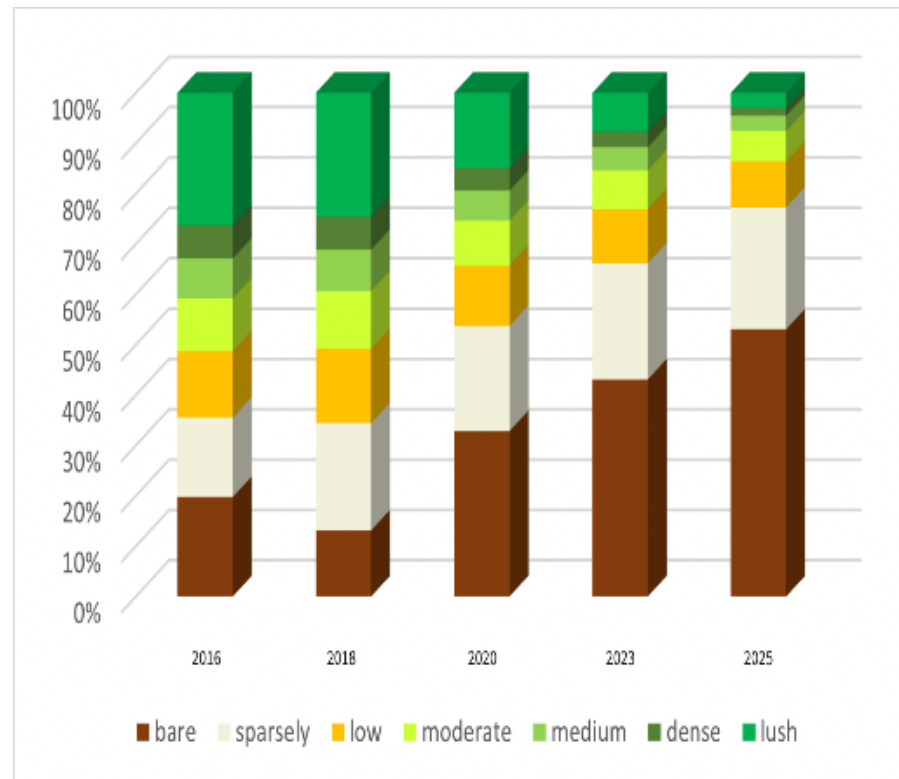
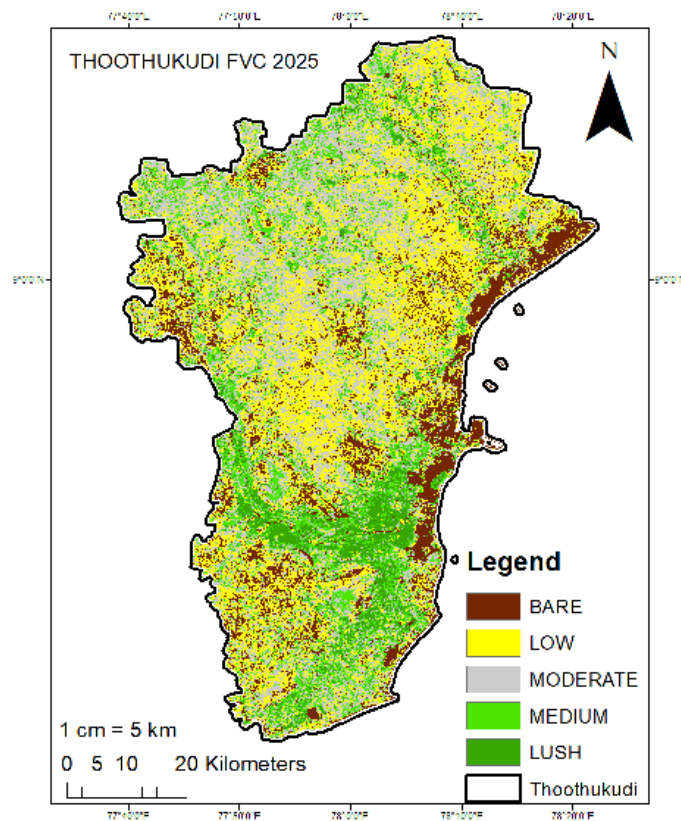


figure 5.5 spatiotemporal analysis of fractional vegetation cover thoothukudi district 2016 – 2025

5.7 Land Use Land Cover Analysis

The Land Use Land Cover (LULC) analysis provides an overview of the spatial distribution of different land use categories in Thoothukudi . The results indicate that built-up and residential areas dominate the landscape, accounting for approximately 40–50% of the total area. Vegetation and agricultural lands contribute around 30–40%, while water bodies and other categories occupy less than 10–15%.

The spatial distribution map shows that built-up areas are concentrated in urban centers, while vegetation is distributed in agricultural regions and scattered patches. The bar graph supports this distribution by providing a clear quantitative representation of land use categories.

The variation in LULC is a key factor influencing all other indices. Areas with high built-up density correspond to higher temperatures, stronger urban heat island effects, and increased ecological stress. In contrast, vegetation and water bodies contribute to cooling and environmental stability. The results highlight the impact of land use changes on thermal and ecological conditions. The dominance of built-up areas emphasizes the need for balanced land use planning to ensure sustainable development.

To mitigate the adverse environmental impacts of rapid urbanization, it is imperative to integrate these LULC-derived insights into the district's master planning framework. The analysis demonstrates that unchecked sprawl into agricultural and scrub

forest areas is not only consuming productive land but is also the primary driver behind localized thermal escalation.

A primary strategic recommendation is the implementation of "Informed Zoning Regulations." By designating mandatory green buffer zones between industrial clusters and residential settlements, the district can create natural barriers to heat propagation and improve local air quality. Furthermore, the results necessitate a transition toward compact urban development models that prioritize vertical growth over outward expansion, thereby preserving the ecological fringes that provide essential cooling services.

Finally, the temporal methodology employed in this study highlights the efficacy of using remote sensing for continuous land-use monitoring. It is recommended that municipal planning authorities institutionalize periodic LULC assessments using GIS and satellite-derived data. This would enable planners to monitor land-use trajectories in real-time, assess the effectiveness of current environmental policies, and make data-driven adjustments to development projects. By fostering a growth trajectory that recognizes the intrinsic value of vegetation and water bodies, the district can effectively mitigate the expansion of thermal hotspots and secure a healthier, climate-resilient future for its inhabitants.

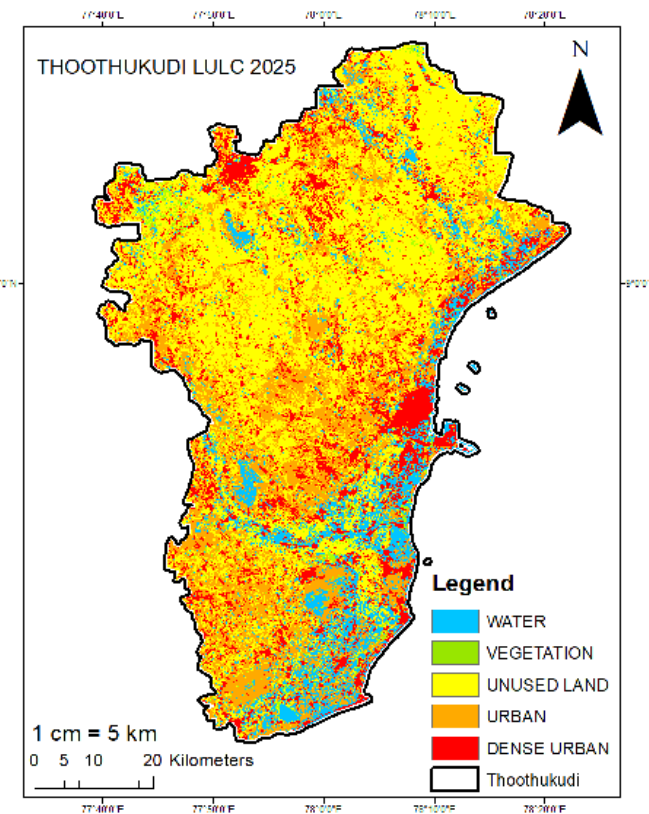
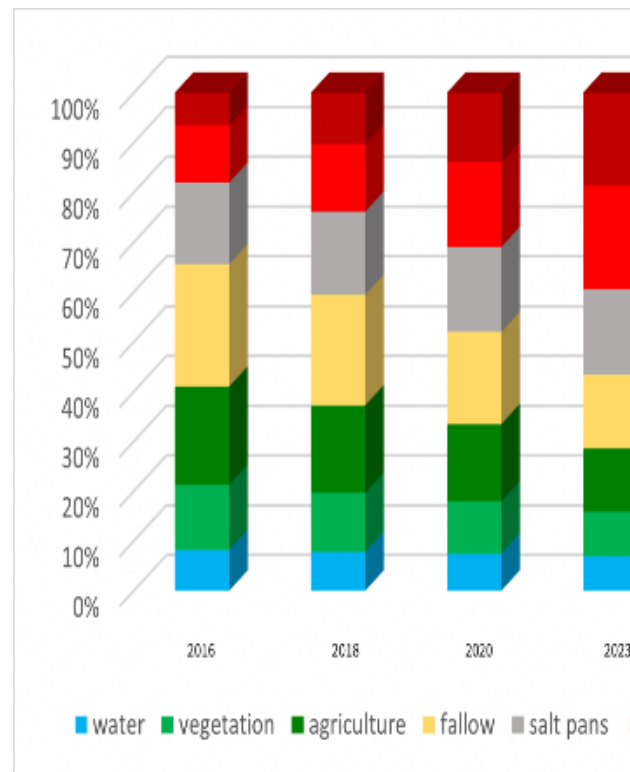
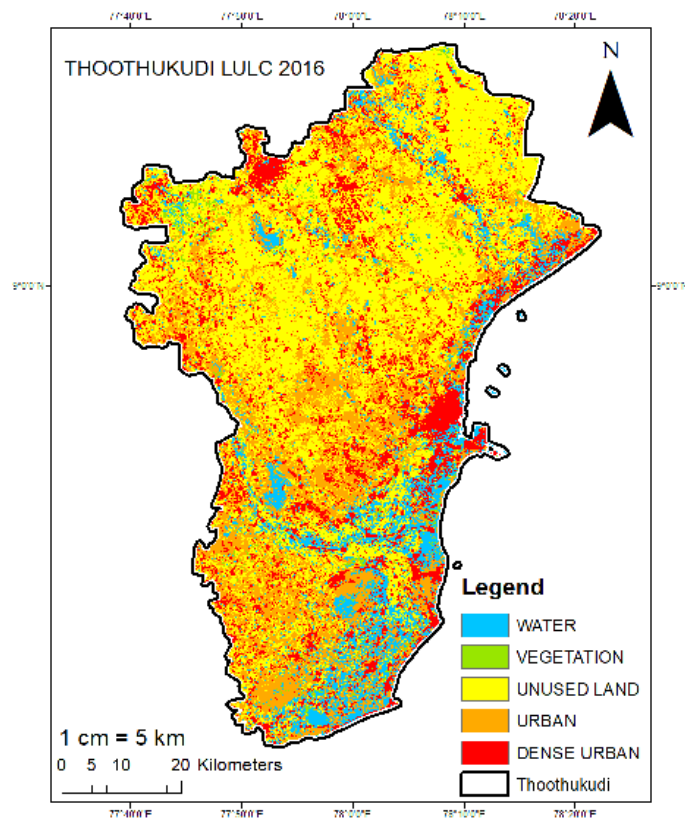


figure 5.6 spatiotemporal analysis of land use and land cover thoothukudi district 2016 - 2025

5.8 Urban Heat Island Analysis

The Urban Heat Island (UHI) analysis reveals the extent of temperature variation between urban and non-urban areas in Thoothukudi. The results indicate that warm to hot categories together account for approximately 50–60% of the study area, while extreme heat zones contribute around 10–15%. Mild and neutral categories form a comparatively smaller proportion of about 25–30%.

The spatial distribution map shows that high UHI intensity is concentrated in urban and industrial regions, where built-up surfaces dominate. In contrast, areas with vegetation and proximity to the coast exhibit lower UHI intensity. The bar graph supports these observations by providing a quantitative representation of the distribution of UHI categories.

The variation in UHI intensity is primarily influenced by land use patterns. Built-up surfaces such as concrete and asphalt absorb heat during the day and release it slowly, leading to higher temperatures.

The lack of vegetation reduces cooling through evapotranspiration, further intensifying the heat island effect. Although the coastal location of Thoothukudi provides some cooling through sea breeze, it is not sufficient to counterbalance the effects of urbanization. The results clearly demonstrate that urban expansion is a major factor contributing to increased thermal stress, highlighting the need for sustainable urban planning strategies.

First, the integration of "Green Infrastructure" is critical. Increasing the density of urban green spaces and tree-lined avenues can significantly enhance local evapotranspiration, providing a natural cooling mechanism that directly offsets the heat absorbed by concrete and industrial structures. Strategic reforestation, particularly in areas transitioning from agricultural to industrial use, can serve as a vital thermal buffer.

Second, policy interventions should focus on the adoption of "Cool Roof" technologies and high-albedo materials for industrial rooftops. As identified in this study, these large, low-reflectivity surfaces act as primary heat sources; retrofitting them with reflective coatings can drastically reduce solar heat absorption.

Finally, urban master planning must prioritize "Ventilation Corridors." By optimizing building density and orientation, the city can allow the natural sea breeze to penetrate deeper into urbanized zones, effectively flushing out accumulated thermal energy. By integrating these thermal insights into the district's development framework, Thoothukudi can transition toward a model that balances necessary industrial expansion with the maintenance of public thermal comfort and ecological health.

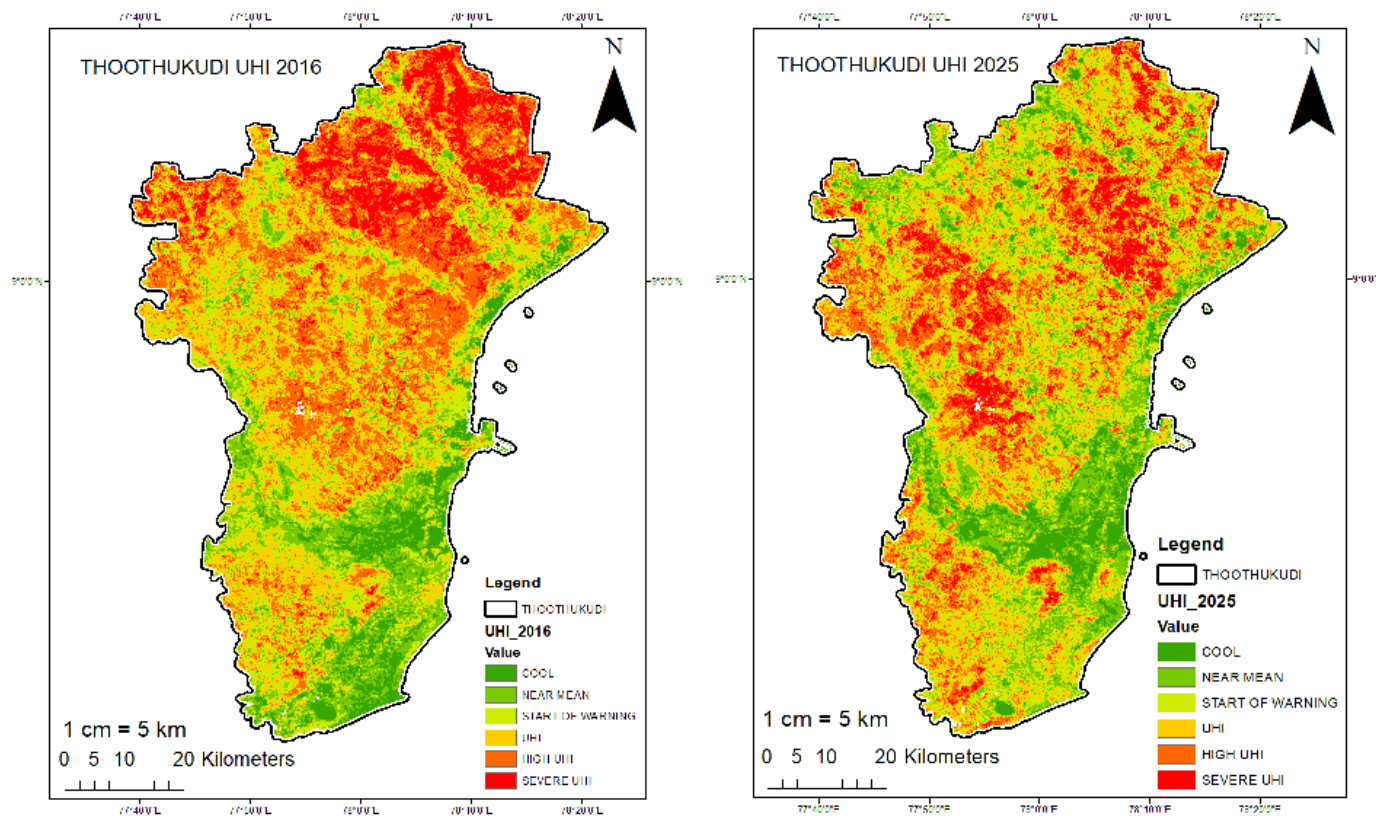


figure 5.7 spatiotemporal analysis of urban heat island thoothukudi district 2016 - 2025

5.9 Urban Thermal Field Variance Index Analysis

The Urban Thermal Field Variance Index (UTFVI) is used to assess the ecological condition of the study area based on thermal characteristics. The results for Thoothukudi indicate that normal to bad ecological conditions dominate, covering approximately 50–60% of the area. Poor to severe categories account for around 20–25%, while only 10–15% of the region falls under good or excellent conditions.

The spatial map shows that areas with poor ecological conditions are concentrated in regions with high Land Surface Temperature and dense built-up structures. In contrast, areas with better ecological conditions are associated with higher vegetation cover and lower temperatures. The bar graph supports this observation by clearly indicating the distribution of ecological categories.

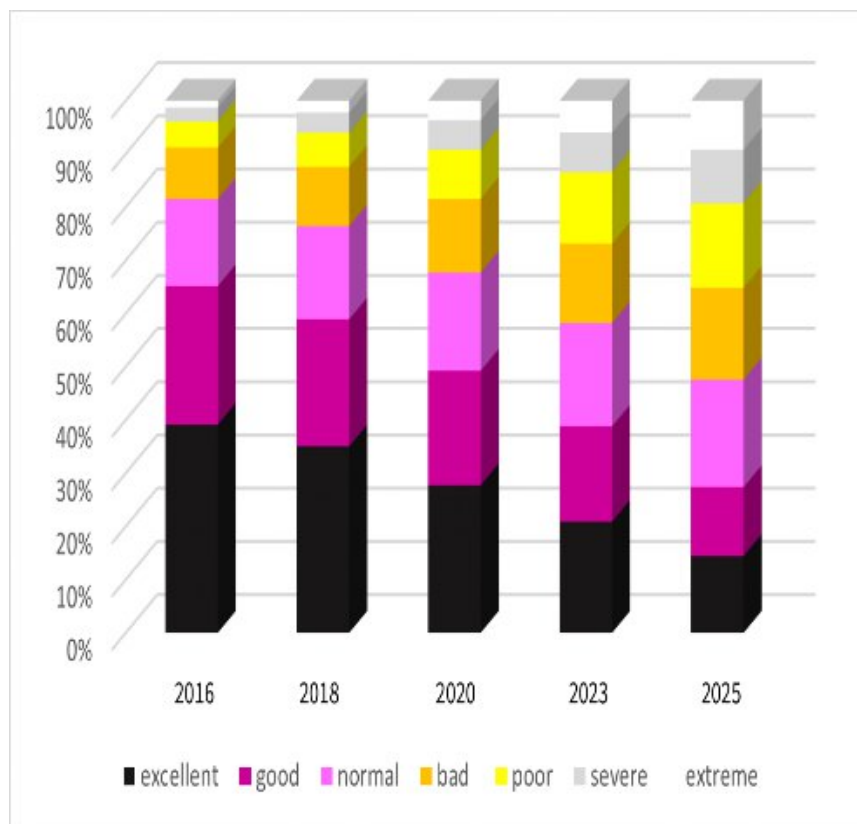
The variation in UTFVI reflects the impact of urbanization and thermal stress on environmental quality. High temperature zones, particularly in urban and industrial areas, contribute to ecological degradation. The limited presence of areas with good ecological condition suggests that the environment is under significant stress. The results emphasize the need for increasing green spaces and implementing environmental management strategies to improve ecological quality. UTFVI thus serves as a critical indicator for evaluating the sustainability of urban environments.

Consequently, these findings provide a necessary evidence base for

urban planners and environmental policymakers to prioritize land-use interventions. Areas identified as having "Bad" or "Severe" ecological quality should be prioritized for immediate mitigation, such as the implementation of urban cooling infrastructure, the restoration of natural water bodies, and the enforcement of stricter zoning laws to limit uncontrolled industrial encroachment in heat-sensitive zones.

Moreover, the strong correlation between high UTFVI values and dense built-up regions demonstrates that current urban expansion patterns in Thoothukudi are directly contributing to long-term ecological instability.

To counter this trend, future development projects must incorporate "Thermal Comfort" as a key performance indicator in project proposals. By integrating these spatial insights into a comprehensive land-use strategy, the district can better balance its rapid industrial ambitions with the essential need for a healthy, habitable, and climate-resilient environment. This multi-scalar approach—linking local thermal indices to broader sustainable development goals—is essential for ensuring the long-term environmental viability of coastal industrial corridors.



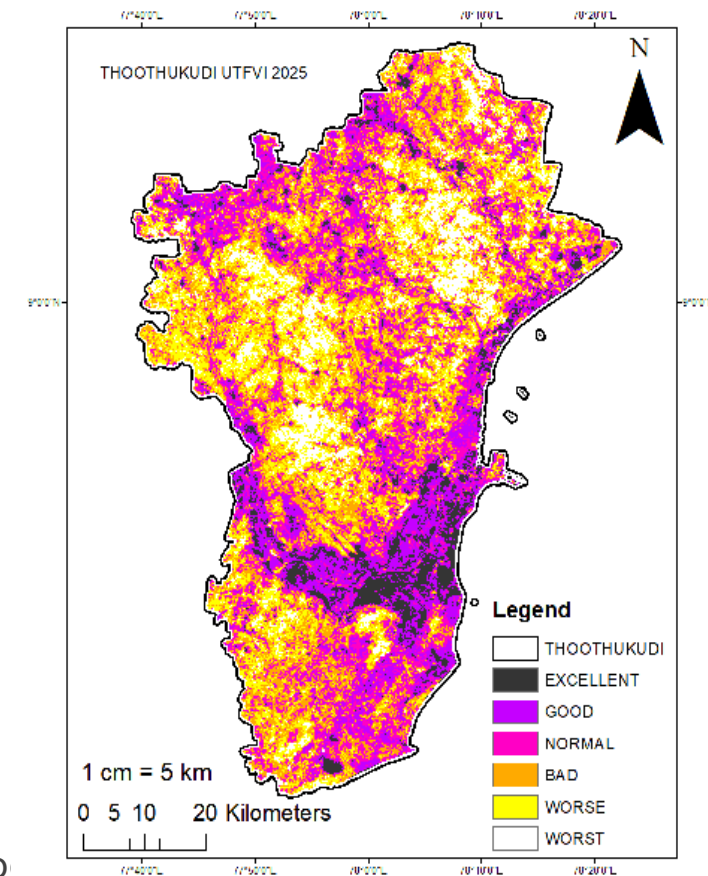
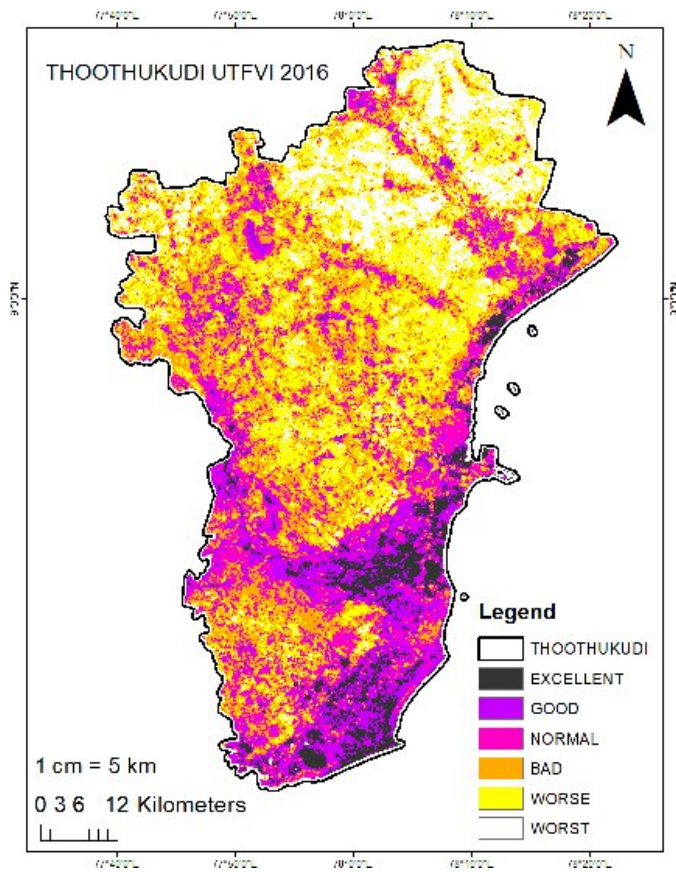


figure 5.8 spatiotemporal analysis of urban thermal field variance index tho

5.10 Crop Water Stress Index Analysis

The Crop Water Stress Index (CWSI) analysis provides insight into the level of water stress experienced by vegetation in Thoothukudi. The results indicate that moderate to high stress conditions affect approximately 40–50% of the area, while extreme stress is observed in about 10–15%. Areas with low or no stress account for approximately 30–35%.

The spatial distribution map shows that higher stress levels are concentrated in regions with high temperature and low vegetation cover, particularly in urban and barren areas. In contrast, areas with vegetation and access to water resources exhibit lower stress levels. The bar graph further supports this distribution by quantifying the proportion of each stress category.

The variation in CWSI is influenced by both thermal conditions and moisture availability. High temperatures increase evapotranspiration, leading to greater water stress, while limited rainfall and saline soil conditions in Thoothukudi further exacerbate the problem. The results highlight the vulnerability of agricultural lands and vegetation to thermal stress. The findings suggest that improved water management and irrigation practices are necessary to reduce stress levels and enhance agricultural productivity.

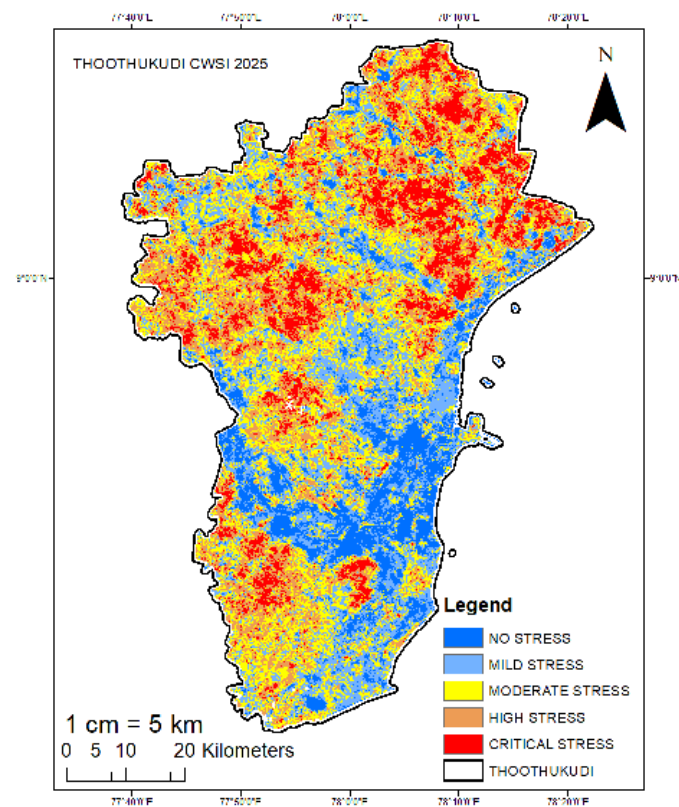
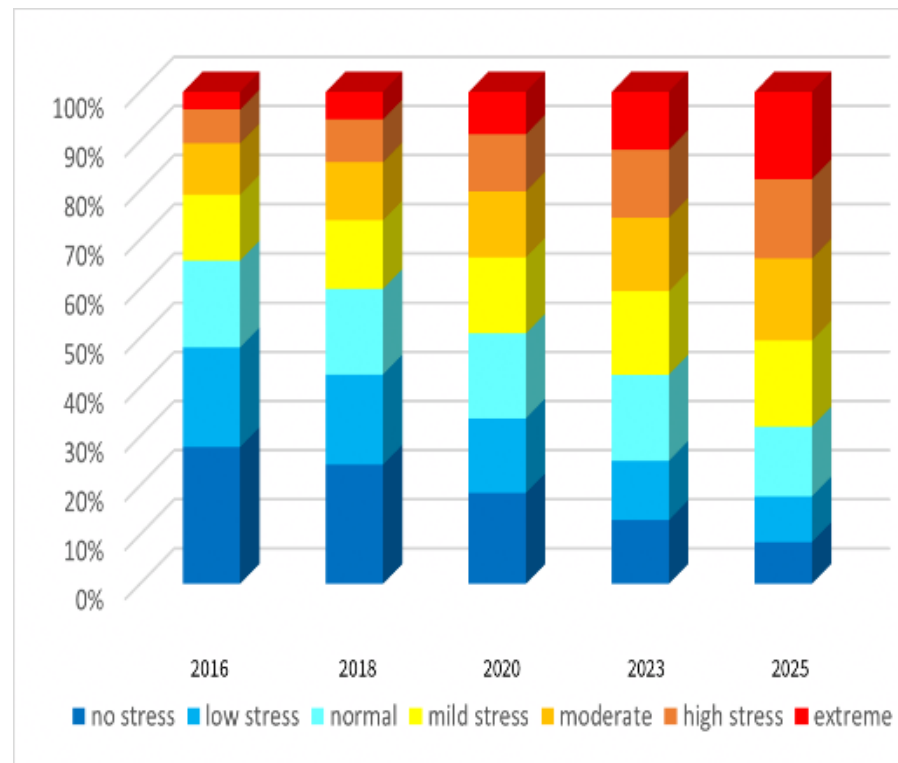
To mitigate these risks and bolster the district's resilience against climate-induced drought, local authorities and agricultural planners must prioritize adaptive water management strategies. Given the

clear spatial correlation between high CWSI values and areas with limited water access, a transition toward precision agriculture is essential.

The implementation of water-efficient technologies—such as drip irrigation systems and real-time soil moisture monitoring—could significantly alleviate water stress in high-vulnerability zones, allowing for optimized crop growth even under thermal pressure.

Furthermore, the findings underscore the critical need for integrated watershed management. Restoring traditional water bodies and expanding rainwater harvesting infrastructure are vital steps to stabilize soil moisture levels and recharge groundwater reserves.

Additionally, aligning cropping patterns with regional moisture availability is crucial. By promoting the cultivation of drought-resistant crop varieties better suited to Thoothukudi's semi-arid conditions and saline soil, the agricultural sector can reduce its reliance on precarious groundwater supplies. Ultimately, integrating CWSI data into the district's broader ecological planning framework allows for a shift from reactive crisis management to proactive, data-driven stewardship of regional water resources, ensuring the long-term sustainability of the district's agrarian economy.



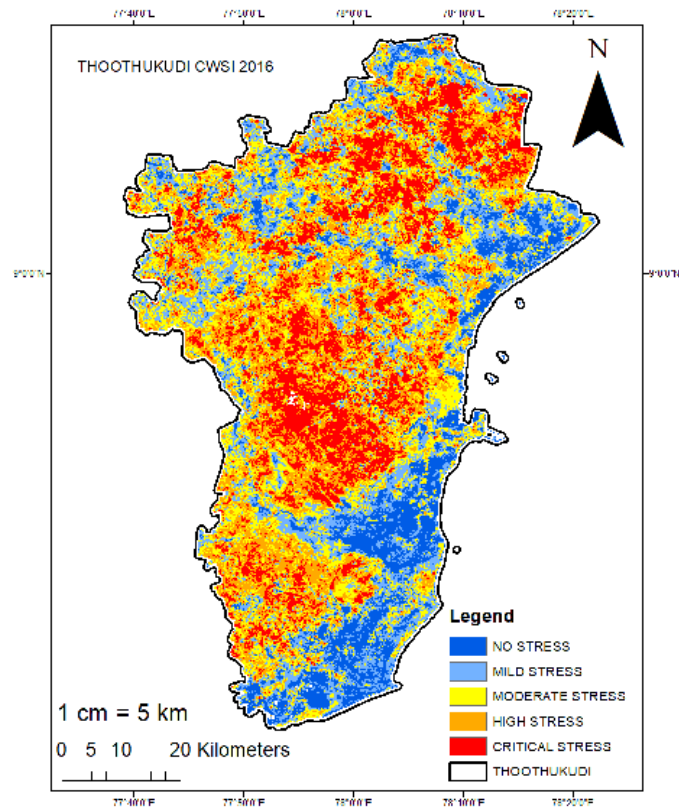


figure 5.9 spatiotemporal analysis of crop water stress index
thoothukudi district 2016 - 2025

5.11 Heat Health Risk Index Analysis

The Heat Health Risk Index (HHRI) analysis evaluates the impact of thermal conditions on human health in Thoothukudi. The results indicate that moderate to warning risk categories dominate, covering approximately 50–60% of the study area. High to severe risk zones account for around 20–25%, while only 10–15% of the area is classified as safe.

The spatial map shows that high-risk areas are concentrated in densely populated urban and industrial regions, where high temperatures and exposure levels are significant. Lower risk areas are associated with vegetation and coastal zones. The bar graph supports these observations by providing a quantitative breakdown of risk categories.

The variation in HHRI is influenced by a combination of thermal intensity, population exposure, and vulnerability. Urban areas with high LST and dense population are more susceptible to heat-related health issues. The limited proportion of safe zones indicates that a large portion of the population is at risk. The results highlight the need for effective heat mitigation strategies, such as increasing green cover and improving urban planning, to reduce health risks.

To operationalize these findings, the municipal administration should develop an integrated Heat Action Plan (HAP) tailored specifically to the district's demographic and climatic profile. This involves establishing early warning systems that utilize real-time temperature and humidity data to predict extreme heat events,

allowing for timely public health advisories and the deployment of emergency services in high-risk zones.

Simultaneously, structural interventions are required to alleviate the heat burden on the most vulnerable populations. This includes retrofitting industrial colonies and residential clusters with "cool roof" technologies, as well as expanding urban green spaces that serve as critical cooling centers during peak summer months. By prioritizing these structural adjustments, the district can systematically lower the population's exposure level, thereby improving the overall HHRI score.

Ultimately, this HHRI analysis serves as a vital decision-support tool for evidence-based policy. By mapping health risks against demographic vulnerability, the study enables targeted resource allocation, ensuring that mitigation efforts are concentrated in the areas of greatest need. Incorporating these findings into the district's public health and urban development strategies is a necessary step toward building a climate-resilient future for Thoothukudi, effectively protecting its residents from the intensifying threat of heat-related illnesses and ensuring the socio-economic stability of the coastal region.

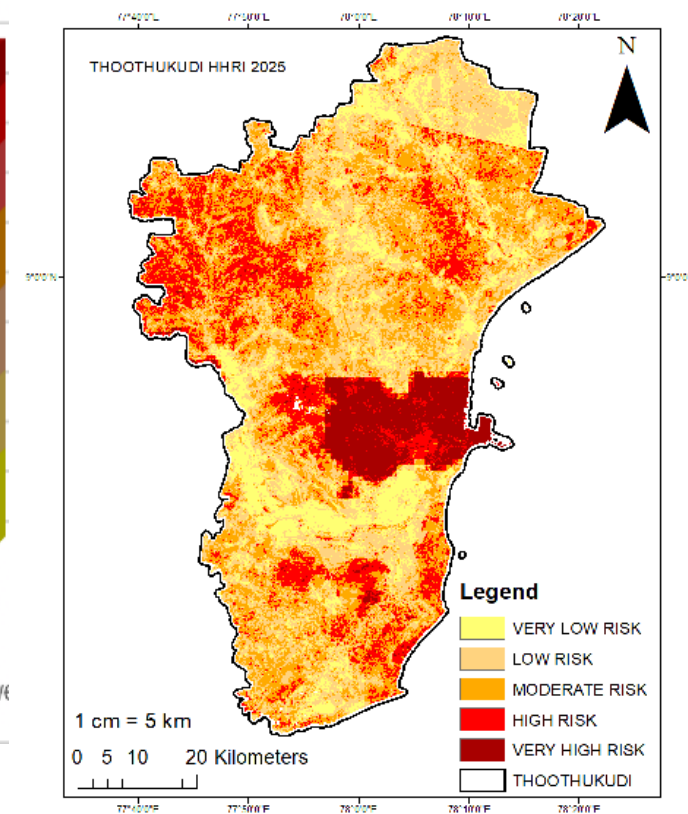
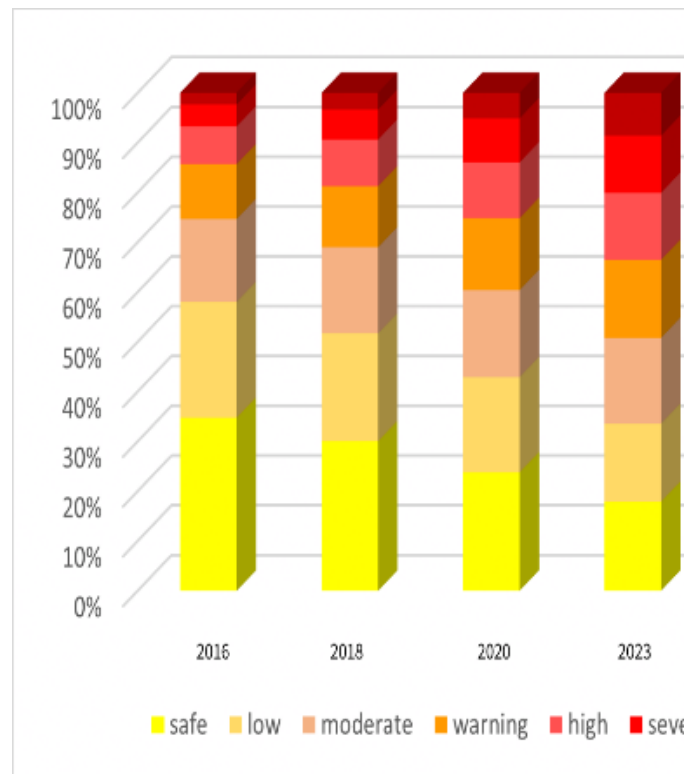
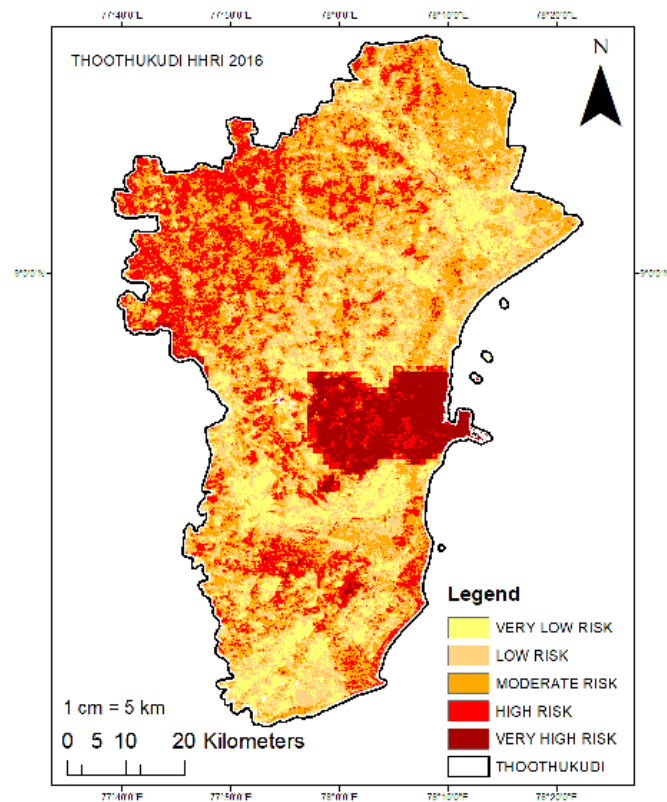


figure 5.10 spatiotemporal analysis of heat health risk index Thoothukudi district 2016 - 2025

CHAPTER 6

SUMMARY AND CONCLUSION

6.1 SUMMARY

The environmental landscape of Thoothukudi is currently navigating a profound and systemic biophysical transformation. This research provides a comprehensive, multi-dimensional diagnostic of the district's ecological health, revealing a complex and increasingly precarious interplay between aggressive land-use changes, thermal intensity, and long-term climatic resilience. By synthesizing advanced remote sensing indices through the high-performance Google Earth Engine (GEE) platform, this study offers a holistic overview of how a decade of rapid urbanization and unchecked industrial growth has fundamentally altered the region's surface energy balance and ecological integrity.

The most striking finding of this investigation is the sheer scale of the environmental shift: a staggering 40% to 60% of the study area is now categorized by moderate to high levels of thermal intensity and ecological stress. This spatial dominance of "heat zones" suggests that the environmental equilibrium of this coastal district is reaching a critical tipping point. The results indicate that the natural cooling mechanisms of the region are being

systematically dismantled, with direct and severe implications for public health, energy consumption, and the long-term stability of local ecosystem services. The research effectively maps a "thermal landscape" where the traditional boundaries between industrial zones and residential safety are increasingly blurred by radiating surface heat.

The data uncovers a stark, binary contrast between the "concrete heat traps" of the built environment and the dwindling natural landscapes. Urban and industrial clusters are identified as the primary engines of elevated Land Surface Temperature (LST) and the subsequent intensification of the Urban Heat Island (UHI) effect. The research demonstrates that the physical blueprint of the city—specifically the replacement of porous, moisture-retentive soil with heat-absorbing materials like concrete, bitumen, and industrial roofing—has created a self-sustaining thermal legacy. Indices such as the Urban Thermal Field Variance Index (UTFVI) and the Heat Hazard Risk Index (HHRI) show a near-perfect correlation with high-density construction. In these identified thermal hotspots, the absence of pervious surfaces and the loss of convective cooling through evapotranspiration prevent natural temperature regulation, resulting in localized zones of extreme heat hazard that pose a direct threat to the urban "livability index."

Furthermore, by integrating the Crop Water Stress Index (CWSI) with biophysical markers, this study proves that thermal stress is not

merely a matter of human discomfort—it is a hydrological and agricultural crisis. The research reveals that elevated surface temperatures are accelerating soil moisture depletion and increasing the water demand of the remaining urban flora, creating a feedback loop of ecological degradation. The methodological robustness of this research is a key highlight, as the multi-index satellite analysis successfully captured the multi-dimensional nature of urban heat with high statistical reliability.

6.2 CONCLUSION

The research conducted on the spatio-temporal thermal transformation of Thoothukudi confirms that the region has undergone a significant environmental transition during the period from 2016 to 2025, largely influenced by rapid urbanization and land-use dynamics. The research objectives were successfully achieved through systematic analysis and interpretation of multi-temporal geospatial data.

The retrieval and processing of decadal satellite data were effectively carried out using Landsat-8 TIRS and OLI sensors within the Google Earth Engine (GEE) platform. Cloud-free annual composites ensured temporal consistency and high accuracy, enabling reliable monitoring of environmental changes across the study area.

The quantification of thermal and biophysical indices established

a strong relationship between land-use changes and temperature variations. The results indicate that the built-up area has increased significantly from 312.45 km² (18.6%) in 2016 to 498.72 km² (29.8%) in 2025, marking an expansion of nearly 186.27 km². In contrast, vegetation cover has declined from 654.38 km² (38.9%) to 472.16 km² (28.2%), reflecting a loss of approximately 182.22 km². Open and fallow lands have also reduced due to their conversion into urban and industrial spaces. This shift from pervious to impervious surfaces has resulted in a noticeable increase in Land Surface Temperature (LST), with mean LST rising from 29.4°C in 2016 to 33.1°C in 2025, and peak temperatures exceeding 36°C in dense urban zones.

The analysis of Urban Heat Island (UHI) intensity and Urban Thermal Field Variance Index (UTFVI) indicates that approximately 52.7% of the total geographical area falls under moderate to high thermal stress categories. Industrial corridors, port areas, and high-density residential zones have emerged as dominant thermal hotspots. Meanwhile, cooler zones associated with vegetation and water bodies have diminished spatially, reducing their overall cooling impact.

The ecological risk modeling using Heat Hazard Risk Index (HHRI) and Crop Water Stress Index (CWSI) highlights the growing vulnerability of both natural and human systems. The findings reveal that nearly 46.3% (≈ 775 km²) of vegetation areas are under moderate

to severe water stress conditions. Additionally, about 41.8% of the urban population zones fall within high thermal risk categories, indicating increased susceptibility to heat-related health issues and ecological degradation.

The applied methodology was validated through statistical measures, demonstrating high reliability and accuracy for coastal thermal studies. Based on these findings, the study proposes key mitigation strategies, including the development of urban green corridors, afforestation programs, regulation of industrial expansion, and sustainable water resource management practices.

In conclusion, Thoothukudi has experienced a marked increase in built-up area and a corresponding decline in vegetation and open spaces over the past decade. This land transformation has directly intensified land surface temperature and strengthened urban heat island effects. If these trends continue, the region may face severe environmental and socio-economic challenges. Therefore, immediate implementation of sustainable urban planning and climate-resilient strategies is essential to ensure long-term ecological balance and urban sustainability.